



Mapping the greenwashing research landscape: a theoretical and field analysis

Canio Forlano¹ · Enrico Battisti¹ · Paola de Bernardi¹ · Tomáš Klietík²

Received: 10 April 2024 / Accepted: 3 February 2025 / Published online: 24 February 2025
© The Author(s) 2025

Abstract

This study provides a comprehensive overview of greenwashing research in business and management, focusing on its conceptual and theoretical foundations. Through a systematic literature review of 97 peer-reviewed articles from Scopus and Web of Science, we use bibliometric analysis to map the evolution of greenwashing studies and network analysis and identify four thematic clusters: (1) symbolic management and CSR communication, (2) environmental regulations and institutional complexity, (3) performance and sustainable practices, and (4) marketing, perception, and trust. Additionally, we perform a full-text analysis to map the theoretical frameworks employed across these clusters, revealing the dominance of macro-level theories (e.g., legitimacy, institutional, and stakeholder theories) and their cross-level interactions with meso- and micro-level approaches. By developing a multi-level theoretical framework, we provide a structured approach to understanding greenwashing across different analytical layers. Then, focusing on the theories used to investigate greenwashing, we identify some avenues for future research. This study has practical implications for understanding how greenwashing affects corporate governance, financial performance, and stakeholder trust, providing insights for enhancing transparency and accountability in sustainable business practices.

Keywords Greenwashing · Corporate social responsibility · Symbolic management · Systematic literature review · Bibliometric analysis · Network analysis

JEL Classification D21 · L2 · L5 · M00 · Q5

1 Introduction

In an era where environmental consciousness is increasingly at the forefront of public discourse, companies are under mounting pressure to adhere to eco-friendly practices (Santos et al. 2023; Szabo and Webster 2021). This pressure is part of a broader expectation for corporate social responsibility (CSR), a concept that encompasses a company's voluntary actions to integrate social, environmental, and economic concerns into their business operations and stakeholder interactions (Yang et al. 2024). Concurrently, consumers are becoming more discerning, increasingly scrutinizing the authenticity of corporate eco-friendly claims (Chatterjee et al. 2024). Sensing the potential repercussions for their brand integrity and social legitimacy, some organizations could create exaggerated or misleading portrayals of their environmental initiatives (Heras-Saizarbitoria et al. 2020; Shuang et al. 2024).

In response to these pressures and incentives for CSR, a growing number of firms are engaging in “CSR washing” or decoupling—that is, engaging in symbolic communication about their CSR initiatives without making substantive changes to their organizational structure and practices (Talpur et al. 2023). Originating in neo-institutional theory, the concept of decoupling refers to when an organization seeks to gain legitimacy by adopting a façade of acquiescence to widely held societal norms while making—whether intentionally or not—minimal commensurate changes to internal practices (Ali Gull et al. 2023; Graafland and Smid 2019). As such, CSR scholars have extensively used decoupling to conceptualize a divergence between CSR communication and CSR practice (Pope and Wæraas 2016). This strategy, commonly known as “greenwashing” when applied to environmental claims, aims to project an environmentally responsible facade that may not necessarily reflect the organization's actual environmental conduct (Vangeli et al. 2023). The term was coined by environmentalist Jay Westerveld in the 1980s in response to his personal experience staying at a hotel that encouraged guests to reuse towels for ecological reasons while lacking a genuine environmental strategy (Motavalli 2011). This concept, inspired by “whitewashing”—the practice of concealing or diminishing negative information—pertains specifically to sustainability. So, although a consistent academic consensus on the definition of greenwashing remains elusive (Torelli et al. 2020), it generally describes the act of misleadingly portraying products, services, or corporate policies as environmentally friendly, or overstating their ecological advantages (de Freitas Netto et al. 2020).

As highlighted in a recent report from RepRisk (2023), one of the world's largest ESG science companies, there was a 35% increase in greenwashing incidents globally in 2023 compared to 2022. The comprehensive analysis conducted by RepRisk, encompassing data from more than 100,000 public and private entities across 165 countries (while excluding corporate self-disclosures), identified the oil and gas sector as the primary contributor to this trend, followed by the financial services and banking industries. These latter sectors experienced a 70% surge in greenwashing incidents, reflecting their growing involvement in these practices. Additionally, the report found that 18% of the companies engaged in greenwashing since 2018 have also participated in “social washing”. This global analysis underscores the pervasive

nature of greenwashing across various industries and geographical regions, highlighting the increasing entanglement of environmental and social issues in CSR practices.

Given this background, the terms greenwashing, CSR washing, and CSR decoupling have gained significant attention in academic circles, particularly in the marketing and management domains. This includes investigations into public responses to greenwashing events and communication (Blazkova et al. 2023; Liu et al. 2023), as well as brand trust, consumer behavior, and purchase intentions (Nyilasy et al. 2014; Vangeli et al. 2023). More recently, a broader range of disciplines has examined the strategic decisions and organizational mechanisms that spur greenwashing (Kim and Lyon 2015; Lyon and Maxwell 2011), alongside the role of accounting, environmental disclosure (Barbu et al. 2022; De Meyst et al. 2023; Sciulli and Adhariani 2023), and “green finance,” which links environmental claims to financial strategies and investment decisions (Bandini et al. 2022; Deschryver and de Mariz 2020; Tuhkanen and Vulturius 2022). In addition, efforts to quantify greenwashing at the firm level have emerged, introducing indicators that measure the gap between a company’s perceived and actual environmental performance (Dorffleitner and Utz 2023). This broadened focus has led to various studies aimed at consolidating greenwashing research. Such work often includes bibliometric analyses to chart and scrutinize relevant scholarly work, providing overviews of field trends and summaries of past research findings (Pendse et al. 2023; Wang et al. 2023). Researchers have also examined the basic principles and categories of greenwashing (de Freitas Netto et al. 2020), its links with stakeholder management (Santos et al. 2023), and its origins, classifications, and impacts (Yang et al. 2020). Lastly, efforts have been made to explore greenwashing as it relates to specific sectors or contexts, like the agri-food industry (Montero-Navarro et al. 2021) and supply chain management (Inês et al. 2023).

However, none of these systematic literature reviews (SLRs) have investigated which theories have informed studies related to greenwashing. As a result, several scholars have emphasized the need for theoretical investigations (Bernini and La Rosa 2023; Gatti et al. 2019; Schmuck et al. 2018; Wagner et al. 2020) since theories offer more than descriptive insights; they provide analytical frameworks that explain how greenwashing practices interact with various factors, offering a comprehensive view of this complex phenomenon. Equally important is the identification of key thematic clusters within the literature. These clusters highlight focal areas and expose both well-explored and under-researched aspects of greenwashing (de Freitas Netto et al. 2020). Understanding how theories and themes interact is critical for addressing the gaps in the literature and providing actionable insights into greenwashing practices, offering practical implications for both academics and practitioners. Hence, our work extends prior reviews by exploring the intersections between thematic clusters and their underlying conceptual foundations, as well as the theories used to investigate greenwashing. We identify areas where theoretical contributions are lacking and suggest opportunities for cross-fertilization. This systematic review, therefore, bridges existing gaps and provides a more integrated understanding of how greenwashing theories align with thematic clusters. Accordingly, our study addresses four research questions:

RQ1 How has the business and management literature addressing greenwashing evolved over time?

RQ2 How have different theoretical perspectives informed the distinct thematic clusters within the greenwashing literature?

RQ3 What are the prevalent theoretical perspectives in greenwashing studies, and how do they interrelate?

RQ4 How can future studies address existing gaps in greenwashing research?

To answer these research questions, our examination progresses through different phases. First, we outline the methodology and review protocol employed in this study. We then present the descriptive results of the final sample, consisting of 97 articles from Scopus and Web of Science, selected based on our inclusion and exclusion criteria. Next, based on a keywords co-occurrence analysis and an in-depth qualitative review of the papers' full texts, we discuss the four thematic clusters that emerge: (1) symbolic management and CSR communication, (2) environmental regulations and institutional complexity, (3) performance and sustainable practices, and (4) marketing, perception, and trust. In the fourth phase, we combine this qualitative thematic analysis with network analysis to highlight the key theories and theoretical frameworks within greenwashing studies and develop recommendations to advance research in the field of greenwashing. Finally, we delineate the study's main limitations and further developments.

This comprehensive approach not only contributes to the academic discourse but also has significant implications for understanding and addressing greenwashing practices in real-world contexts. The investigation of greenwashing from a theoretical perspective is crucial for developing a comprehensive understanding of this complex phenomenon. This structured approach enables researchers to navigate the complex landscape of greenwashing studies more effectively, and examining the theories used in greenwashing research helps to uncover the underlying assumptions and explanatory mechanisms proposed by different scholars. Hence, this theoretical lens can reveal how various research domains conceptualize and explain greenwashing, potentially leading to a more nuanced and multifaceted understanding of the phenomenon. The implications of this theoretical analysis also extend beyond academia, as a solid theoretical foundation can inform more effective strategies for identifying, preventing, and addressing greenwashing practices in real-world contexts. By synthesizing different theories and examining their alignment with key themes, we aim to facilitate the application of theoretical frameworks at different levels of analysis in greenwashing research.

2 Theoretical background

Over the years, the concept of greenwashing has evolved into a multifaceted phenomenon situated at the intersection of corporate communication, environmental responsibility, and stakeholder perceptions. Initially framed as disinformation propagated by organizations to shape their public image, it encapsulated the essence of misleading environmental claims aimed at building a more favorable brand perception (Laufer 2003). Subsequent research broadened this view, with Delmas and Burbano (2011) characterizing greenwashing as the intersection of poor environmental performance and positive communication about environmental performance. This definition highlights the critical discrepancy between a company's actions and its communications, a key aspect of greenwashing. Moreover, Delmas and Burbano's (2011) seminal paper offered a comprehensive analysis of the multilevel drivers behind greenwashing, which can be categorized into external (e.g., market forces and regulatory contexts), organizational (e.g., firm characteristics, organizational culture, incentive structures), and individual (e.g., cognitive biases and decision-making mechanisms) factors. Their work laid the foundation for a more nuanced understanding of the drivers and implications of greenwashing at different levels of analysis. Matejek and Gössling (2014) further dissected the concept, viewing greenwashing as a company's deliberate manipulation of environmental communications; they noted that symbolic actions could entirely overshadow substantial environmental efforts. Meanwhile, Lyon and Maxwell (2011) emphasized the strategic nature of greenwashing, viewed as messages intentionally crafted by companies to create a more favorable environmental image than their actual practices warrant. This strategic aspect is further explored with signaling theory, which has been instrumental in explaining how companies use green claims as a means of conveying (misleading) information about their environmental performance to stakeholders (Kim and Lyon 2015).

In this regard, another important contribution to the conceptualization of greenwashing has come from Mahoney et al. (2013); they highlighted that greenwashing can involve selective positive disclosure, where companies emphasize favorable environmental outcomes to impress stakeholders, even if not entirely true. Accordingly, Guo et al. (2017) emphasized that greenwashing often arises as firms attempt to balance pressures from different stakeholder groups, leading to selective environmental disclosure as a strategic choice aimed at mitigating reputational risks and enhancing legitimacy in the eyes of stakeholders. However, Marquis et al. (2016) challenged this view, finding that firms with poor environmental performance are less likely to engage in selective disclosure. Indeed, their research suggested that poor environmental practices increase visibility and push firms toward transparency, particularly in countries with stronger scrutiny and global norms. Two main drivers of greenwashing emerge from this perspective, rooted in the intersection of institutional and stakeholder theories (Ferrón-Vílchez et al. 2021): a proactive approach, where greenwashing is driven by internal strategic decisions to gain legitimacy, and a reactive approach, which responds to external pressures such as societal norms, regulations, and expectations that shape organizational behavior toward environmental responsibility.

In this vein, another evolution in the understanding of greenwashing is reflected in Seele and Gatti's (2017) accusatory-based definition, which frames greenwashing as a co-created external accusation based on perceived misleading environmental claims. Arguing that greenwashing is largely shaped by stakeholder perceptions, they highlight the relational dynamics between organizations and external parties, further underscoring the importance of external perceptions in defining and identifying greenwashing practices. However, besides these conceptualizations, most researchers rely on definitions of greenwashing from authoritative sources like the *Oxford English Dictionary* and Greenpeace, which define it as deliberate corporate actions aimed at misleading stakeholders about environmental performance (Zioło et al. 2024). The debate around the intentionality of greenwashing is significant, with scholars like Nyilasy et al. (2014) and Bowen and Aragon-Correa (2014) arguing that it should only be considered greenwashing when there is clear intent to deceive. In this regard, Szabo and Webster (2021) suggest that companies with a history of greenwashing need to transition to truthful practices to rebuild trust, potentially using "anti-impact" reports that acknowledge sustainability gaps. For unintentional greenwashers, understanding environmental impacts across the supply chain and seeking additional ways to differentiate—beyond marketing—is critical.

These diverse perspectives reflect the multidisciplinary nature of greenwashing and the challenge of establishing a unified definition, as greenwashing manifests at various organizational levels, from product-level claims to firm-wide sustainability communications. Despite efforts to categorize its forms, the complexity of corporate communication, environmental challenges, and regulatory variations underscore the need for a strong theoretical foundation to effectively understand and address greenwashing. While various theories have been applied, ranging from institutional theory to stakeholder, legitimacy, and signaling theories, this study aims to develop a more cohesive and integrated framework of these perspectives to provide a more comprehensive understanding of greenwashing. Indeed, emerging forms such as "greenwashing" (unrealistic environmental promises) and "greenhushing" (undercommunication of environmental initiatives, a practice also referred to as "brown-washing") have introduced additional dimensions to the problem (Seele and Gatti 2017; Testa et al. 2018b). Moreover, as environmental concerns increasingly influence corporate strategy and public discourse, the multifaceted nature of greenwashing calls for exploring how the different themes related to greenwashing have informed the different theories used to investigate the phenomenon (Bernini and La Rosa 2023; Schmuck et al. 2018).

In addition, recent research has further expanded the conceptual framework of greenwashing by situating it within the broader context of CSR decoupling, which occurs when there is a misalignment between a company's public commitments or reports on sustainability (the "talk") and its actual performance or implementation of these commitments (the "walk") (Amores-Salvadó et al. 2023; Tashman et al. 2019). In this framework, greenwashing represents a positive gap in CSR decoupling, where external communication exceeds internal action. As a result, several scholars have recognized that a more nuanced understanding of greenwashing must consider the broader concept of CSR decoupling or CSR washing (Talpur et al. 2023; Velte 2023). For example, de Freitas Netto et al. (2020) acknowledged that not considering terms

such as CSR washing and decoupling represented a limitation of their study, influencing the number of articles selected in their SLR. A similar limitation was found in previous bibliometric studies and SLRs on the topic (Santos et al. 2023; Wang et al. 2023; Yang et al. 2020). This study addresses these gaps by mapping the alignment between theoretical frameworks and key thematic clusters in greenwashing literature. By identifying the theories that have informed greenwashing research, we provide insights into how future studies can build on these foundations to address existing gaps and offer new avenues for cross-fertilization across disciplines.

3 Methodology

As Tranfield et al. (2003) highlight, SLRs are foundational in consolidating knowledge within a research field and in presenting the current state of the art and identifying gaps to direct future research. Systematic reviews are thus pivotal in expanding and refining the understanding of a research area, providing depth and context (Sauer and Seuring 2023). They involve a thorough evaluation of literature grounded in well-defined research problems and a methodical, critical examination of published studies. This approach aims to surpass the limitations of conventional narrative reviews by implementing comprehensive, unbiased, and transparent steps in all research phases, from sampling to data collection and reporting (Christofi et al. 2017). The replicable nature of SLRs ensures robust, valid, and broadly applicable findings, fostering academic discourse, guiding public policy, and enhancing stakeholder awareness (Paul et al. 2021). In this study, we follow the four structured steps outlined by Kraus et al. (2020): planning the review (as delineated within the introduction), identifying and evaluating studies, extracting and synthesizing data, and disseminating the review findings. This approach ensures transparency, replicability, and verifiability, maintaining the integrity of the research (Baima et al. 2020; Vrontis et al. 2022).

In the first step of our data analysis, we performed a bibliometric analysis of sample papers through the R-package Bibliometrix (Aria and Cuccurullo 2017), which allowed us to generate a variety of visual representations of our bibliometric data through its powerful data visualization capabilities. This tool is increasingly popular among management scholars for analyzing bibliographic databases (Forliano et al. 2021; Martín-Navarro et al. 2023).

The second step involved using VOSviewer software for conducting network analysis and visualizing clusters through keyword co-occurrence analysis, paired with a full-text review of the papers to enable a thematic investigation (Kraus et al. 2022; Öztürk et al. 2024). As done in previous SLRs for more thorough and repeatable outcomes than what could be achieved through manual analysis (De Bernardi et al. 2021; Schöbel et al. 2024), this step also included normalizing the dataset of raw keywords using OpenRefine (<https://openrefine.org>). OpenRefine is an open-source software designed to help clean up large datasets; it offers several algorithms and clustering methods for identifying similar keywords in terms of spelling or phonetics (Verborgh and De Wilde 2013), accounting for variations such as different spellings, singular versus plural forms, British versus American English, use of hyphens,

and other similar factors. It is important to note that OpenRefine does not automatically merge these similar terms but instead suggests potential matches to be manually approved. Hence, we proceeded accordingly, ensuring that we maintained control over the whole data cleaning process. This process was necessary to ensure consistency in keywords that essentially meant the same thing but were represented differently in the dataset (e.g., “greenwashing,” “green-washing,” “green washing”). After this initial round of computer-assisted cleaning, we performed an additional manual review to address variations in terminology that OpenRefine could not identify. For example, we manually reconciled instances in which acronyms were used, such as “Corporate Social Responsibility” and “CSR” or “Environmental, Social, and Governance” and “ESG.” This normalization process is crucial for keyword co-occurrence analysis, as network nodes depend on keyword frequencies, and inconsistencies could result in similar keywords appearing in different clusters and potentially compromise the thematic analysis.

3.1 Data collection

To fulfill our research objectives, we began by selecting the two most prominent bibliographic databases in social sciences for gathering raw data (Battisti et al. 2023; Bertello et al. 2023): Web of Science (WoS) by Clarivate Analytics and Scopus by Elsevier. WoS encompasses over 15,000 journals and more than 90 million documents, while Scopus indexes upwards of 20,000 active titles, including peer-reviewed journals, books, and conference proceedings, amounting to around 69 million records. Following previous bibliometric studies and SLRs on the subject (e.g., Moodaley and Telukdarie 2023; Talpur et al. 2023; Vangeli et al. 2023), we developed a search string to capture studies investigating greenwashing. In addition, our keyword selection was informed by the evolving understanding of greenwashing within the broader context of CSR practices, as discussed in our theoretical background. Hence, we aimed to address the limitations noted in previous SLRs, which acknowledged that not considering terms such as CSR washing and decoupling may have restricted their scope (de Freitas Netto et al. 2020; Santos et al. 2023; Wang et al. 2023; Yang et al. 2020). While this search string included CSR-related terms, our study maintained a central focus on greenwashing. The inclusion of these additional terms was designed to capture studies that discuss greenwashing within the context of CSR decoupling or CSR washing so that we could include research on broader misleading CSR communications and the misalignment between CSR communication and environmental practices, as recent literature has situated greenwashing within this broader framework. However, our subsequent screening process ensured that only papers with greenwashing as a central theme were included in the final analysis. The Scopus search string was as follows: TITLE-ABS-KEY (greenwash* OR ((csr OR "corporate social responsibilit*") W/3 (wash* OR decoup*))). We used the Boolean operator “OR” to combine these keywords and wildcards and capture all the potential combinations of specific words. Moreover, to enhance the comprehensiveness and precision of the results (Buchkremer et al. 2019), we employed the proximity operator W/3 in Scopus and its equivalent NEAR/3 in Web of Science, allowing for a targeted search where two keywords must appear within three words of each other. As shown in Fig. 1, the

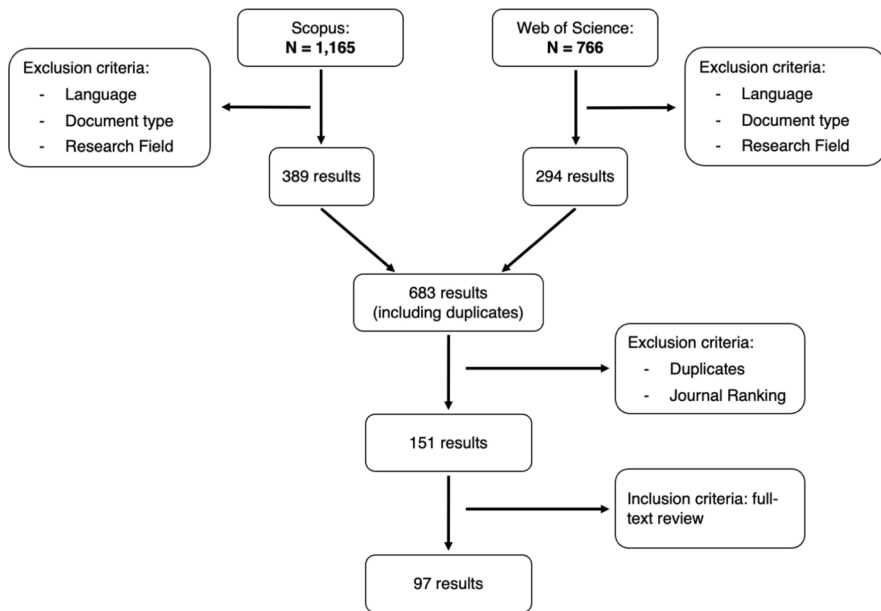


Fig. 1 Overall research design. Source: Own elaboration

initial search (conducted in September 2023) yielded 1,165 results in Scopus and 766 results in WoS. To refine our dataset, we subsequently applied some inclusion and exclusion criteria, as detailed in the following paragraph.

3.2 Inclusion and exclusion criteria

In our investigation of greenwashing, language limitations led us to consider only English-language articles (Follmer and Jones 2018). Second, like other systematic reviews, we sought only articles published in scholarly peer-reviewed journals, regarded as containing the most influential and validated knowledge (Keupp et al. 2012). Thus, we excluded books, book chapters, editorials, working papers, and conference proceedings. Third, applying the relevant categories in Scopus and WoS, we further refined the sample by focusing on the articles' research fields. Hence, we included only those published in the business and management domains. This led to 389 articles in Scopus and 294 in WoS. After removing duplicates, further refinement was conducted by selecting studies from journals ranked 3 or above in the latest Academic Journal Guide (Chartered Association of Business Schools 2021). This ranking is highly regarded in business and management studies, and, as shown in Kraus et al. (2020), it is comparable with other criteria, like the JCR Impact Factors and the German VHB-JOURQUAL3 (JQ3). Hence, following previous SLRs that adopted the same approach (Buchkremer et al. 2019; Kossyva et al. 2023), we could focus on the most pertinent and current knowledge on the topic while keeping the sample size manageable.

After downloading the remaining 151 articles, we implemented a rigorous two-stage screening process. In the first stage, each researcher independently scrutinized the titles and abstracts of all articles. We included articles that focused primarily on greenwashing in a business or management context and contributed to understanding the evolution, theoretical foundations, or practical implications of greenwashing. In cases of uncertainty, we proceeded to a second stage involving a full-text review. During this stage, we assessed whether the article addressed greenwashing as a central theme rather than merely mentioning it, and whether it offered substantive discussion or implications regarding greenwashing. Importantly, we did not exclude articles based on the absence of explicit theoretical references. This decision was made to ensure that we captured a full spectrum of greenwashing research, including practice-led studies that implicitly inform theoretical advancements. By including these articles, we aimed to provide a more comprehensive analysis that encompasses both conceptual and applied perspectives, thus enriching our mapping of the field. This approach resulted in the selection of 97 articles.

4 Descriptive results of research on greenwashing

In this section, we present a bibliometric analysis conducted as part of our SLR on greenwashing. This analysis utilizes various performance indicators to address the primary research question of our study: *RQ1. How has the business and management literature addressing greenwashing evolved over time?* We thus undertook a comprehensive descriptive examination of our sample of research papers to trace the evolution of articles over time, identify the most frequently cited articles, determine the most prolific journals and countries, and recognize the authors with the highest productivity and influence in this field.

As shown in Fig. 2, the analysis of article trends within our timeframe (i.e., 2003–2023) reveals a relatively low and stable number of articles published until 2012, fluctuating between 1 and 3 articles per year. This suggests that greenwashing was not a primary focus of research or public debate during those years. Since 2013, there has been a notable increase in the number of articles addressing greenwashing, likely in response to various international regulations and guidelines aimed at curbing such practices, such as the 2012 update of the United States' Green Guides and Australia's 2011 introduction of provisions against misleading environmental claims. However, the greatest surge in scholarly interest is observed after 2015. Indeed, the notable uptick in sustainability interest following the establishment of the Sustainable Development Goals (SDGs) and the Paris Agreement in 2015 led to increased environmental commitments from various sectors. Subsequently, the legitimacy of these pledges increasingly came under scrutiny, emphasizing the need for truthful and non-deceptive claims. Furthermore, another significant peak in research interest is evident post-2019, likely driven by renewed interest due to the European Green Deal. Advocating for sustainable growth and stricter regulation of companies' environmental information, this policy led the European Commission to adopt a proposal for a directive on green claims in March 2023.

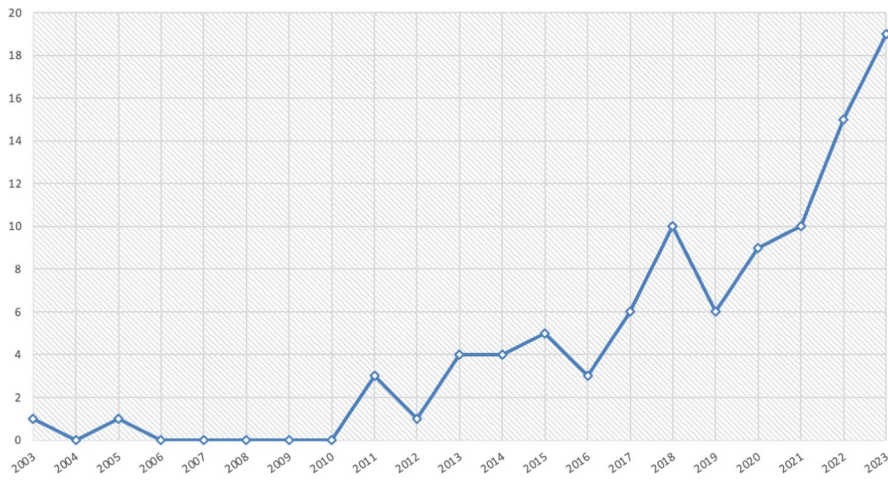


Fig. 2 Distribution of publications over time. Source: Own elaboration

To assess the most impactful research in the field of greenwashing, we categorized articles based on the citations they received (Table 1). Citations often synthesize the influence of a publication into a single metric, providing a robust measure for identifying key scholarly contributions (Merigó et al. 2016). In particular, to avoid penalizing more recent articles, Table 1 ranks the 15 most influential articles based on the average number of citations received per year (TC/Y) instead of total citations (TC).

As shown in Table 1, the three most influential articles in our sample are Delmas and Burbano (2011), Szabo and Webster (2021), and Marquis et al. (2016). Distinguishing between environmental, organizational, and individual-level drivers of greenwashing, Delmas and Burbano (2011) received the highest number of yearly citations and total citations (TC/Y 69.38, TC 902). The second most cited article is Szabo and Webster (2021) (TC/Y 53.67, TC 161), which employed a mixed-method approach to examine the implications of green marketing and perceived greenwashing, offering insights into consumer perceptions and company challenges in environmental communication. Notably, this study introduced a methodological innovation by using neurophysiological measurements, marking a departure from traditional self-reported surveys to study consumer responses to greenwashing. Marquis et al. (2016), the third most cited article (TC/Y 45.25, TC 362), challenged previous assumptions about selective disclosure practices in firms with poor environmental performance, contributing to our understanding of how institutional pressures influence corporate environmental transparency. These highly cited works represent significant contributions to the field, shaping our understanding of greenwashing from organizational, consumer, and institutional perspectives.

Although the *Journal of Business Ethics* appears 6 times among the 15 most influential papers (see Table 1), Table 2 shows that the journal that publishes the most articles on greenwashing is *Business Strategy and the Environment* (22), followed by the *Journal of Business Ethics* (21) and *Business and Society* (4). This distribution suggests a comprehensive investigation of greenwashing across multiple dimensions

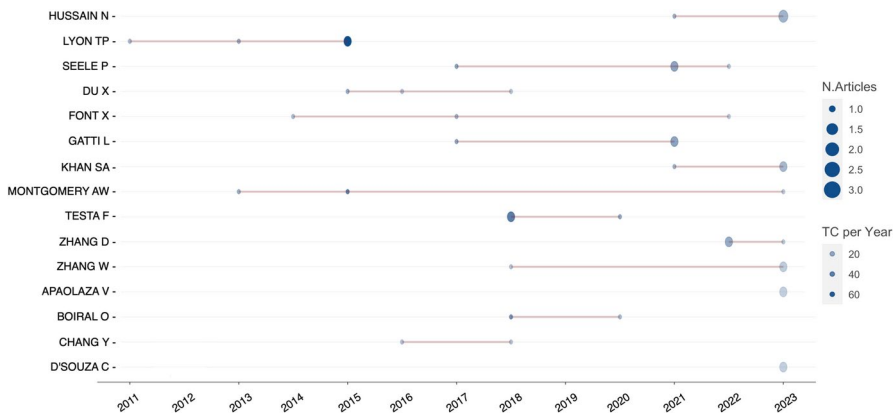
Table 1 Citation analysis for the top 15 papers in the dataset, sorted by average yearly citations received (TC/Y). Source: Own elaboration

#	Author	Year	Title	Journal	TC/Y	TC
1	Delmas and Burbano	2011	The Drivers of Greenwashing	Calif. Manag. Rev	69.38	902
2	Szabo and Webster	2021	Perceived Greenwashing: The Effects of Green Marketing on Environmental and Product Perceptions	J. Bus. Ethics	53.67	161
3	Marquis et al	2016	Scrutiny, Norms, and Selective Disclosure: A Global Study of Greenwashing	Organ. Sci	45.25	362
4	Lyon and Montgomery	2015	The Means and End of Greenwash	Organ. Environ	44.22	398
5	Chen and Chang	2013	Greenwash and Green Trust: The Mediation Effects of Green Consumer Confusion and Green Perceived Risk	J. Bus. Ethics	43.27	476
6	Wu and Shen	2013	Corporate Social Responsibility in the Banking Industry: Motives and Financial Performance	J. Bank. Financ	38.55	424
7	Testa et al	2018	Internalization of Environmental Practices and Institutional Complexity: Can Stakeholders Pressures Encourage Greenwashing?	J. Bus. Ethics	37.50	225
8	Laufer	2003	Social Accountability and Corporate Greenwashing	J. Bus. Ethics	34.95	734
9	Siano et al	2017	“More Than Words”: Expanding the Taxonomy of Greenwashing After the Volkswagen Scandal	J. Bus. Res	33.43	234
10	Parguel et al	2011	How Sustainability Ratings Might Deter ‘Greenwashing’: A Closer Look at Ethical Corporate Communication	J. Bus. Ethics	32.62	424
11	Torelli et al	2020	Greenwashing and Environmental Communication: Effects on Stakeholders’ Perceptions	Bus. Strat. Environ	31.25	125
12	Lee and Raschke	2023	Stakeholder Legitimacy in Firm Greening and Financial Performance: What About Greenwashing Temptations?	J. Bus. Res	31.00	31
13	Nyilasy et al	2014	Perceived Greenwashing: The Interactive Effects of Green Advertising and Corporate Environmental Performance on Consumer Reactions	J. Bus. Ethics	30.30	303
14	Mahoney et al	2013	A research Note on Standalone Corporate Social Responsibility Reports: Signaling or Greenwashing?	Crit. Perspect. Account	29.36	323
15	Kim and Lyon	2015	Greenwash vs Brownwash: Exaggeration and Undue Modesty in Corporate Sustainability Disclosure	Organ. Sci	28.33	255

of business and ethics. While *Business Strategy and the Environment* leads, indicating a strong focus on the strategic implications of greenwashing for businesses and their environmental policies, the *Journal of Business Ethics* closely follows, highlighting the growing concern and scrutiny over the ethical aspects of greenwashing in corporate practices. Moreover, the presence of journals with a more specialized focus shows that greenwashing is not just a critical issue in business strategy and ethics but also impacts a wide range of business functions and sectors. It reflects an expanding

Table 2 Citation analysis for the top 15 journals in the data-set, sorted by total number of publications (TP). Source: Own elaboration

#	Journal	TP
1	Business Strategy and the Environment	22
2	Journal of Business Ethics	21
3	Business and Society	4
4	International Review of Financial Analysis	3
5	Journal of Advertising	3
6	Journal of Business Research	3
7	Organization and Environment	3
8	Industrial Marketing Management	2
9	International Journal of Contemporary Hospitality Management	2
10	International Journal of Hospitality Management	2
11	Journal of Sustainable Tourism	2
12	Management Science	2
13	Organization Science	2
14	Tourism Management	2
15	Transportation Research Part E: Logistics and Transportation Review	2
	Others	22
	Total	97

**Fig. 3** Top 15 authors' production over time. *Note* The size of the circle correlates with the number of articles the author has published that year. The color's intensity indicates the annual citations received. Source: Biblioshiny, based on the dataset (colour figure online)

recognition of greenwashing's implications across different business domains, from marketing and finance to organizational behavior and tourism management—reflecting a broader scope in recent years.

Table 1 highlighted the most cited articles on greenwashing, showcasing their influence in the field, while Table 2 presented the most productive journals, indicating their contribution to greenwashing research. Impact and productivity are both key metrics in academic research. In Fig. 3, these metrics are combined to assess the significance of authors in the domain of greenwashing over time. Productivity is

work, the centrality of a node (i.e., a specific keyword) is determined by the frequency of its co-occurrences with other keywords. The strength of the link between any two keywords is defined by how often they are jointly mentioned in scholarly works. The size of a node in the network directly correlates with the frequency of the keyword's occurrence, with larger nodes representing more commonly used keywords. In our analysis, we only included keywords that appeared at least twice, resulting in a network of 82 keywords.

As Fig. 4 shows, our analysis reveals four distinct clusters within the greenwashing literature. The first cluster, illustrated in red (17 keywords), predominantly concentrates on themes related to symbolic management and CSR communication. The second cluster, colored in green (29 keywords), primarily addresses environmental regulations and institutional complexity. The third cluster, depicted in blue (18 keywords), focuses on performance and sustainable business practices. Lastly, the fourth cluster, shown in yellow (18 keywords), encompasses keywords that pertain to marketing, perception, and trust. The following sub-sections provide a thorough discussion of each theme that emerged based on our full-text analysis of the papers in the sample.

5.1 Red cluster: symbolic management and CSR communication

This cluster, characterized by terms like “agency theory,” “attribution theory,” “CSR communication,” “symbolic management,” and “transparency,” delves into the ethical dimensions of corporate behavior in the domain of greenwashing. The focus is primarily at the organizational level, examining how companies strategically manage their environmental image and communicate their CSR efforts.

5.1.1 Conceptual foundations

At the core of this cluster lies the concept of symbolic management, as explored by Matejek and Gössling (2014). Symbolic management refers to the strategic use of symbols, narratives, and organizational actions to shape stakeholders' perceptions, particularly in terms of a company's environmental and social responsibility. In the context of greenwashing, symbolic management often manifests as companies portraying themselves as environmentally conscious through various symbolic acts or communications without taking substantive actions to support these claims. This creates a facade of environmental responsibility, potentially misleading stakeholders and the public while serving as a tool for improperly gaining legitimacy (Torelli et al. 2020). As noted in their conceptual comparison between greenwashing and “machine-washing”—that is, the unethical use of artificial intelligence to provide misleading information, conceal wrongdoings, or circumvent socio-environmental regulations—Seele and Schultz (2022) further stressed the potential (bad) implications of such novel solutions in reinforcing symbolic management practices.

Hence, the concept of legitimacy emerges as a crucial element in this cluster, closely tied to symbolic management practices and CSR communication. Within the organizational context, legitimacy refers to the perceived appropriateness and acceptability of an entity's actions within a socially constructed system of norms, values,

and beliefs. In this context, greenwashing can be seen as an attempt to manipulate perceptions of legitimacy without making substantive changes to environmentally harmful practices (Hassan et al. 2020). This theoretical link helps explain why companies might engage in greenwashing even when it poses reputational risks, as a strategy to gain, maintain, or repair their legitimacy.

5.1.2 Organizational strategies and practices

Several studies have examined how companies strategically utilize symbolic management in their CSR communication to maintain legitimacy. For example, Montgomery et al. (2023) discuss ExxonMobil's CSR communication strategies as a multifaceted example of symbolic management in the oil and gas sector. While the company prominently highlights its biofuel research in public communications, historical evidence reveals that it previously invested in commercials questioning the reality of climate change. Internal documents, however, indicated little doubt about the matter. This discrepancy between public communications and internal knowledge exemplifies a form of strategic misrepresentation to manage public perceptions and maintain legitimacy, further underscored by ExxonMobil's simultaneous financial contributions to climate research institutions. Additionally, its messaging prominently features algae biofuels even though they are only a minor part of its operations—potentially overshadowing its primary focus on fossil fuels (Marshall et al. 2023).

Similarly, the fast fashion industry provides a compelling illustration of similar practices. H&M, a major player in this sector, has publicly committed to utilizing only recyclable or sustainable materials by 2030. This ambitious pledge can be interpreted as a strategic move to enhance the company's legitimacy in a market increasingly concerned with environmental issues. However, the complexity and global nature of the company's supply chain, characterized by several small-scale operations in various countries, presents substantial obstacles to the practical implementation of this pledge, as many suppliers may lack the capacity to comply with these rigorous sustainability standards (Ye et al. 2022). In the technology sector, Amazon's pledge to achieve carbon neutrality by 2040 has attracted significant attention but also criticism. While the goal is ambitious, critics contend that Amazon's rapid expansion, particularly in delivery services and data center operations, may exacerbate its carbon footprint in the short term. Additionally, Amazon's support for legislators denying climate change and initiatives aimed at accelerating fossil fuel extraction further complicates its environmental narrative (Robertson et al. 2023).

Reflecting on brownwashing, Kim and Lyon (2015) pointed out similar results. Brownwashing is a deceptive communication practice that involves strategically downplaying a company's CSR initiatives to avoid appearing overly costly or inconsistent with shareholder interests (Testa et al. 2018b). In some cases, such a strategic approach leads to the complete omission of communications related to CSR initiatives, a practice called greenhushing, such as in the case of big corporations like IKEA not disclosing the usage of certified cotton in its products (Ginder et al. 2021). This can be seen as the antithesis of greenwashing, which entails exaggerating or fabricating CSR efforts to enhance a company's image and legitimacy in the eyes of stakeholders. This manipulation of a corporate image, aimed at maintaining or

enhancing legitimacy, can be particularly deceptive because it exploits the public's assumption that no disclosure of environmental impacts is better than manipulating CSR communication. These examples underscore the intricate and occasionally contradictory nature of corporate strategies employed to secure and sustain legitimacy through environmental commitments—sometimes selectively promoting environmentally positive initiatives, other times spreading potentially misleading information or directing long-term pledges that may conflict with short-term actions.

5.1.3 Organizational antecedents and consequences

Understanding these dynamics also requires an examination of internal resources and capabilities, as emphasized by the resource-based view (RBV) of the firm. The RBV suggests that a company's ability to effectively implement environmental strategies depends on its internal endowments, including tangible and intangible assets such as know-how, corporate culture, and unique capabilities (Hart and Dowell 2011). These resources determine the potential for a corporate strategy to move beyond symbolic gestures and yield genuine environmental improvements (Orazalin et al. 2023). In the context of greenwashing, the RBV provides insight into why some firms might opt for symbolic environmental management as a cost-effective alternative to genuine environmental improvements. Companies with limited resources may view symbolic gestures as sufficient for maintaining legitimacy and competitive advantage, even when they fall short of implementing substantial environmental changes. On the other hand, a "win-win" scenario can be created, where adopting proactive environmental practices not only addresses sustainability concerns but also enhances corporate effectiveness (Testa et al. 2018a). These practices can indeed lead to economic advantages such as cost reductions through more efficient resource use, decreased material waste, and energy savings. Moreover, these practices can also drive innovation, resulting in new processes, products, or technologies that strengthen the company's competitive position or unique capabilities that are difficult for competitors to replicate.

Building on these foundations, agency theory provides crucial insights into the internal dynamics that can lead to greenwashing behaviors. This theory addresses the conflicts that may arise between corporate managers (agents) and shareholders (principals) in sustainability decision making. In the context of greenwashing or brownwashing, it is often utilized to analyze how decisions regarding sustainability are influenced by conflicts between stakeholder interests and corporate executive goals (Seele and Gatti 2017). For instance, managers might engage in greenwashing to project a positive environmental image and satisfy stakeholder demands in the short term, even if this conflicts with long-term shareholder interests or genuine environmental improvements. While primarily focused on individual-level perceptions, attribution theory has important implications at the organizational level as well, bridging the gap between corporate actions and stakeholder interpretations. It helps explain how stakeholders interpret and attribute motives to companies' CSR efforts and environmental claims. In the absence of external information, the theory suggests, consumers are likely to attribute CSR communication to extrinsic motives, such as conforming to prevailing sustainability trends (Gosselt et al. 2019). This theoretical

insight is crucial for understanding how organizational communications strategies can backfire, leading to skepticism and accusations of greenwashing.

5.1.4 Governance and mitigation strategies

Encompassing the rules, practices, and processes by which a company is directed and controlled, different governance and mitigation strategies can be implemented in diminishing symbolic management. First is *transparency and accountability*. While inconsistent CSR messages can lead to confusion and skepticism (Nyilasy et al. 2014; Parguel et al. 2011), transparency is crucial in communicating CSR endeavors (Amin et al. 2022). As Lyon and Montgomery (2015) suggest, the adoption of clear and comprehensive disclosure practices—involving the release of “hard” and verifiable information that can be reliably shared with others—can help build trust with stakeholders and reduce skepticism.

A second strategy consists of *regulatory compliance and voluntary standards*. Companies that voluntarily disclose detailed, verifiable CSR information are less likely to be accused of greenwashing. Moreover, the implementation of robust internal auditing processes and third-party verification of sustainability reports can further enhance credibility and ensure that reported environmental practices reflect actual performance. In this regard, the adoption of established frameworks such as supranational disclosure regulations (Fiechter et al. 2022) or accounting standards like the Global Reporting Initiative (Koseoglu et al. 2021) can further improve such perceptions. Accordingly, the clearer and more open a company is about its CSR activities, the lower the likelihood of being accused of greenwashing.

A third strategy involves *internal oversight and board composition*. A fundamental aspect of governance is the composition of the board of directors. For example, research has shown that the presence of women and independent directors can significantly influence the adoption of sustainable actions, thereby enhancing corporate environmental performance (Eliwa et al. 2023; Koseoglu et al. 2021). The establishment of dedicated CSR committees or compensation schemes also significantly enhances transparency and reducing agency conflicts (Fiechter et al. 2022). These governance mechanisms can help ensure that CSR initiatives are genuinely pursued, rather than merely serving as symbolic gestures. A well-structured board can align managerial actions with the long-term interests of shareholders and other stakeholders, thereby mitigating the risks of greenwashing.

A fourth strategy is *mitigating greenwashing through culture and ethics*. Fostering a corporate culture that prioritizes ethical behavior and sustainability can be a powerful deterrent against greenwashing (Eliwa et al. 2023). Companies that embed environmental values into their corporate culture are more likely to pursue genuine sustainability initiatives rather than resorting to symbolic management. This cultural shift can be supported by leadership commitment, ethical training programs, and incentives that reward sustainable practices (Lee and Raschke 2023).

5.2 Green cluster: environmental regulations and institutional complexity

This cluster explores the intersection between sustainable development and environmental regulations, as well as the role played by different actors in the complex institutional ecosystem to shape the public perception of greenwashing. Keywords such as “consumer behavior,” “decoupling,” “environmental policy,” “environmental disaster,” “stakeholder engagement,” and “sustainable development” emerge for this cluster. While the cluster includes organizational responses, its primary emphasis is on the broader institutional and regulatory frameworks that influence corporate environmental practices and communications.

5.2.1 Conceptual foundations

At the core of this cluster lies the concept of decoupling, which encapsulates the idea of separating economic growth from environmental harm. Decoupling plays a pivotal role in this context, serving as a barometer for the authenticity of corporate environmental practices (Montgomery et al. 2023). It challenges companies to sustain profitability while innovating towards the reconsideration of the environmental impact of long supply chains (Lee et al. 2018a) or the adoption of circular solutions (Testa et al. 2020) to reduce their CO₂ emissions and overall ecological footprint (Bryant et al. 2020).

Institutional theory also emerges as a crucial theoretical foundation in this cluster. It provides a framework for understanding how broader societal norms, regulations, and expectations shape organizational behavior regarding environmental practices (Berrone et al. 2017). This theory helps explain why companies might engage in greenwashing or adopt more genuine environmental practices based on the institutional pressures they face. These pressures—including legal, moral, and societal expectations—significantly influence corporate behavior as organizations strive to align with these norms while balancing traditional business goals with increasing environmental expectations. Simultaneously, stakeholder theory also plays a significant role in this cluster, emphasizing the importance of considering various stakeholder groups in environmental decision making and communication. This theory underscores the evolving nature of stakeholder engagement, which transcends mere information dissemination to become a more dynamic interaction where stakeholders are active participants in environmental discourse and decision making (Testa et al. 2018b; Yamoah et al. 2022). This approach promotes a more integrated perspective on corporate responsibility, highlighting the role of transparency and accountability in enhancing stakeholder trust and cooperation.

5.2.2 Regulatory and institutional frameworks

The focus on decoupling underscores the expectation that responsible businesses must move beyond compliance with current environmental policies, striving instead for continuous improvement in their environmental performance. Several studies have explored the intricate relationship between regulation and corporate behavior. An example of this phenomenon can be observed in the automotive industry, where

some companies have invested in emission reduction technologies to comply with regulations while simultaneously lobbying against stricter standards, demonstrating inconsistency between their public actions and their political influence efforts (Delmas and Burbano 2011). Analyzing US publicly traded firms, Bryant et al. (2020) discussed the paradox of environmental regulatory compliance, which may be essential for environmental protection but could inadvertently jeopardize corporate ambitions by presenting additional opportunity costs that discourage efforts beyond the minimal requirements. At the same time, they found that environmental wrongdoings could be hidden behind actions that appear to go beyond compliance.

Meanwhile, there is also the realm of voluntary actions, such as achieving ISO 14001 certification related to respecting environmental management standards, which can enhance a firm's reputation and demonstrate a commitment to environmental stewardship beyond what is legally required (Heras-Saizarbitoria et al. 2020). Other voluntary initiatives include adherence to the United Nations Global Compact or participation in disclosure programs such as the Carbon Disclosure Project. These voluntary actions can serve as credible signals of a company's environmental commitment, potentially reducing accusations of greenwashing (Bryant et al. 2020). Several studies have thus posited that complementing formal mechanisms of enforcement with informal pressures from public and peer entities can deter or expose greenwashing.

The concept of institutional complexity is particularly relevant in this cluster. Shareholder resolutions, media scrutiny, social media exposure, industry associations, and non-governmental organizations all represent various manifestations of a rich institutional environment, which plays a pivotal role in holding firms accountable and ensuring that their policies and practices align with their professed sustainability commitments (Huang et al. 2020; Ioannou et al. 2023). For instance, environmental NGOs and activist groups are key players in exposing greenwashing and pressuring companies to improve their environmental practices (Gatti et al. 2021). By conducting independent research, publishing reports, and launching public campaigns, these groups highlight gaps between companies' sustainability claims and their actual practices, leading to reputational risks and potential market impacts. Moreover, by leveraging social media and digital platforms, these groups mobilize public opinion, creating grassroots pressure on companies. For example, Strauß et al. (2023) found that news media act as an institutional force that shapes public opinion and sets the agenda on sustainable finance, educating the public and holding companies accountable—thereby pushing them to align more closely with their environmental commitments. Thus, underdeveloped institutional contexts may hinder environmental sustainability and promote companies' selective discourses (Marquis et al. 2016). This is particularly evident in developing countries, where the lack of strong enforcement mechanisms and stakeholder pressure can lead to more widespread and less controlled greenwashing practices (Li et al. 2023a).

Furthermore, institutional complexity arises from the need for multinational companies to navigate different environmental regulations across various jurisdictions, often resulting in inconsistent greenwashing practices or communication (Orazalin et al. 2023). This highlights the need for a coordinated global approach to effectively curb greenwashing. Recent research has also emphasized the crucial role of national culture in shaping environmental responsibility and greenwashing behaviors (Peng

et al. 2023). Indeed, cultural norms act as powerful isomorphic forces, compelling organizations to adopt similar practices and structures based on societal expectations and professional standards. This isomorphism influences corporate decision making, leadership styles, and approaches to environmental ethics. In this regard, Roulet and Touboul (2015) found that firms are more prone to greenwashing in competitive environments, while companies are more likely to take genuine environmental actions in contexts that emphasize individual responsibility. These findings suggest that in contexts where weak government intervention is favored, firms may feel compelled to fill institutional voids through substantive actions, but this tendency is counterbalanced in those environments in which competitive mindsets dominate. This interplay between cultural beliefs and regulatory frameworks underscores the complex ways in which corporate greenwashing practices are shaped across different contexts.

5.2.3 Stakeholder engagement and communication

The evolution of stakeholder engagement from mere information dissemination to active participation in environmental discourse and decision making (Testa et al. 2018b; Yamoah et al. 2022) has profound implications for how companies address greenwashing concerns and manage their environmental communications. This shift is reflected in several regulatory and corporate initiatives. The 2017 Second Shareholder Rights Directive in the EU, for example, was designed to promote long-term shareholder engagement and transparency in corporate governance while reducing short-term risk-taking by managers (Fiechter et al. 2022). In any case, the effectiveness of stakeholder engagement varies significantly among institutional investors. Brandon et al. (2022) found that non-US investors generally achieve better portfolio sustainability scores and improve them after joining responsible investment initiatives, unlike their US counterparts, who often perform no better than non-signatories, even when they claim to fully incorporate sustainability factors. This discrepancy extends to engagement practices, with US investors showing lower levels of engagement compared to non-US investors.

Investors can employ various engagement methods to influence corporate behavior and address greenwashing (Tilba 2022). These include “internal engagement,” where investors influence corporate governance through the appointment of non-executive directors, and “direct engagement,” where large shareholders (often referred to as block-holders) exert influence over senior management. In this sense, informal, behind-closed-doors engagements by these influential investors can be particularly effective. Additionally, collaborative engagement—where multiple investors work together to exert pressure on companies—serves as a strategy to amplify investor voices, especially for smaller shareholders who may not have significant individual influence. Surprisingly, however, Li et al. (2023b) found that when large investors coordinate their actions, it can unexpectedly weaken their influence on corporate governance and increase greenwashing opportunities. This “herd behavior” often leads significant shareholders to ignore their own insights and follow other big investors, obscuring crucial information about a company’s social responsibility performance and making it easier for companies to engage in greenwashing.

As the landscape of stakeholder engagement continues to evolve, new platforms and technologies have emerged as powerful tools for communication and scrutiny. In particular, social media has become a crucial platform for stakeholder engagement, allowing for the rapid dissemination of information but also exposing companies to increased surveillance (Lyon and Montgomery 2013). Consequently, less sophisticated forms of greenwashing are declining due to increased stakeholder vigilance and the spread of new data and monitoring technologies that flatten information asymmetries. Activists now expose greenwashing through inventive methods such as online ratings and categorizing corporate misdeeds. Unilever's experience highlights the complexities of corporate sustainability initiatives. In 1997, Unilever and WWF established the Marine Stewardship Council as an independent non-profit organization focused on safeguarding oceans through a certification program for sustainable fishing (Munasinghe et al. 2021). However, the Council faced criticism for allegedly undermining national fishery management and favoring large commercial fleets, raising concerns that it served Unilever's market interests rather than environmental goals. Similarly, in 2013, Unilever developed a living lab in partnership with *The Guardian* to host a 24-h online dialogue to discuss sustainability challenges, bringing together experts, NGOs, and consumers (Bowen and Aragon-Correa 2014). However, the initiative was met with a Twitter campaign accusing the company of greenwashing (Bowen and Aragon-Correa 2014). These cases illustrate how even partnerships with respected NGOs can be perceived as self-serving rather than genuinely addressing environmental concerns.

The role of social media becomes even more critical during environmental crises, as seen in the 1989 Exxon Valdez oil spill and the 2015 Volkswagen emissions scandal. In the Exxon Valdez disaster, approximately 11 million gallons of crude oil were spilled into the sea after the tanker ran aground near Alaska, causing one of the worst human-made environmental disasters. ExxonMobil's chairman faced widespread criticism for his delayed response, refusal to accept full responsibility, and reliance on passive communication channels like newspaper advertisements. This mishandling significantly amplified public outrage, leading to severe reputational damage, exemplified by around 40,000 customers cutting up their Exxon credit cards in protest (Gilbert et al. 2018). Similarly, when news broke that Volkswagen had been using "defeat devices" to cheat on emissions tests, the company's initial lack of transparency and slow response amplified the scandal's impact. The story spread rapidly across social media platforms, leading to immediate and severe reputational damage, financial consequences, and increased regulatory scrutiny for the company.

These incidents highlight how social media can significantly influence consumer concerns regarding environmental claims (Fernando et al. 2014) and reduce information asymmetry (Li et al. 2023a). Both cases underscore how ineffective stakeholder engagement during an environmental crisis can exacerbate negative outcomes, highlighting the need for prompt, transparent, and sincere communication strategies (Topal et al. 2020). They also demonstrate the evolving landscape of crisis communication—from traditional media in ExxonMobil's case to the rapid, global reach of social media platforms in Volkswagen's situation. These examples emphasize the critical importance of proactive and transparent stakeholder engagement, especially in the context of environmental issues or disasters. Companies must be prepared to

communicate effectively and honestly, leveraging appropriate platforms to promptly address concerns while also being aware of the challenges these platforms present in managing corporate reputation and stakeholder relationships.

5.3 Blue cluster: performance and sustainable business practices

This cluster emphasizes how companies embed sustainability into their core business operations and strategies and the impact of these practices on their overall performance. It explores the relationship between sustainable business practices, financial performance, and corporate strategy. Featuring terms such as “business development,” “environmental performance,” “ESG,” “financial performance,” and “strategic approach,” this cluster centers on the financial analysis and strategic implications of greenwashing. Although some studies found that companies use greenwashing to gain competitive advantage and improve performance (Berrone et al. 2017), this seems to be true especially in under-regulated environments (Delmas and Burbano 2011; Lyon and Montgomery 2015) and in the case of activities targeting external stakeholders with low proximity, such as corporate citizenship (Li et al. 2023a). While the cluster includes various organizational responses to sustainability challenges, its primary emphasis is on how companies integrate environmental and social considerations into their business models and the resulting impacts on performance metrics.

5.3.1 Conceptual foundations

At the core of this cluster lies the concept of integrating sustainability into business strategy and operations. Sustainability is not merely an additional feature for organizations but is increasingly becoming central to companies’ strategic orientations. Indeed, when businesses incorporate sustainability into their core strategy, it often reflects a shift from viewing environmental and social responsibility as cost centers to seeing them as key drivers of long-term value creation (Graafland and Smid 2019). This integration challenges the traditional view of a trade-off between profitability and environmental responsibility, instead positing that these can be mutually reinforcing when properly aligned.

The theoretical perspectives that inform this shift are crucial for understanding greenwashing as a strategic phenomenon. Game theory, for example, offers a lens through which to analyze the strategic decisions that firms make regarding CSR and greenwashing. Wu et al. (2020) introduced a game theory model to differentiate between firms driven by profit and those motivated by genuine social good. Their model highlights that low transparency might lead profit-driven firms to greenwash, while high transparency encourages genuine environmental efforts but reduces extra investments as transparency further increases.

Additionally, in emerging markets where regulations are often weaker, the dynamics of greenwashing differ. Huang et al. (2020) extended the application of game theory to the context of emerging markets, where greenwashing behaviors are notably impeding sustainable progress. Their model explores the competitive pricing strategies of green incumbents (established companies that are genuinely green) and gre-

enwashing entrants (new companies pretending to be green). They found that when the environmental efforts of an incumbent green firm and the greenwashing activities of a new entrant are both low, tolerating greenwashing can sometimes benefit the incumbent by avoiding price competition. Moreover, when the gap between actual and claimed greenness is small, greenwashing can even increase consumer surplus by easing competitive pressures. However, these benefits are context-dependent, and greenwashing is not advantageous beyond specific conditions. The study also highlights how governments in emerging markets may prefer looser enforcement of greenwashing to prioritize economic growth, explaining the prevalence of greenwashing in these regions. Understanding these conceptual foundations helps explain how companies approach sustainability and how these actions translate into various performance metrics. They also offer a framework for analyzing the potential disconnects between stated commitments and actual practices that can lead to accusations of greenwashing.

5.3.2 Greenwashing as a competitive strategy

While often viewed primarily through a communicative lens, greenwashing can also be understood as a deliberate strategic choice with substantial organizational implications. This perspective extends beyond mere “cosmetic” communication, suggesting that greenwashing can drive organizations to take concrete actions in an attempt to align their reality with their public sustainability commitments. Accordingly, this section examines the conditions under which greenwashing is used strategically and the consequences it may have on a company’s market positioning.

In certain environments, particularly those with weak regulatory oversight, companies may find that engaging in greenwashing allows them to enhance their brand image and appeal to environmentally conscious consumers without fully committing to substantive sustainability practices (Berrone et al. 2017). This is particularly true in under-regulated industries where the costs of genuine sustainability initiatives outweigh perceived benefits. Delmas and Burbano (2011) provide evidence that greenwashing is more prevalent in these environments, where companies face fewer repercussions for overstating their environmental achievements. The strategic use of greenwashing is particularly effective in industries where stakeholders such as consumers and investors are distanced from the firm’s actual operations, making it harder to verify claims. For instance, corporate citizenship activities targeting external stakeholders, like community involvement programs, are often more vulnerable to greenwashing (Li et al. 2023a). This vulnerability arises because these initiatives typically receive less scrutiny compared to a company’s operational practices, making it easier for firms to exaggerate their environmental contributions (Kassinis et al. 2022). Unlike core operational processes, which are often subject to rigorous audits and oversight, community-oriented programs are generally seen through a more lenient lens. Stakeholders, including consumers and investors, are less likely to have direct access to verify the actual impact of these programs, leading to potential disconnects between a company’s public claims and its genuine actions. This reduced level of scrutiny allows firms to project a greener image with minimal effort, leverag-

ing these activities as tools for reputation management rather than genuine sustainability efforts.

From a broader perspective, corporate strategies and stakeholder influence are critical in determining whether environmental practices are genuinely adopted or remain symbolic. Siano et al. (2017) note that varying stakeholder expectations can lead to different levels of environmental commitment. When environmental management systems like ISO 14001 are integrated without corresponding behavioral changes, the result can be superficial improvements in environmental management (Heras-Saizarbitoria et al. 2020). This situation often arises when companies prioritize the appearance of compliance over actual environmental outcomes. Testa et al. (2018a) also suggest that stakeholders like shareholders and suppliers, who are invested in the company's long-term sustainability, often promote genuine practices. For example, shareholders might demand transparency and real environmental performance improvements to protect their investments. Similarly, suppliers, who are increasingly interconnected through green supply chain management, support sustainable practices to ensure long-term collaboration.

In contrast, customers can inadvertently drive companies toward greenwashing. This happens because they often lack direct access to information about a company's real environmental practices, leading them to rely on certifications more as marketing tools than as genuine proof of sustainability. Industrial associations may also prioritize economic concerns over environmental goals, sometimes lobbying against stricter regulations, which further complicates the internalization of true sustainability practices. An interesting reflection on organizational implications and micro-level inclinations toward the integration of environmental paradigms comes from Yamoah et al. (2022). Investigating UK food companies, they found that managers are generally aware of environmental issues but often face a misalignment with organizational goals, as the immediate costs of implementing sustainable practices deter proactive changes. This discrepancy can lead to greenwashing, where companies make superficial environmental commitments influenced by public opinion rather than genuine action. This issue was further confirmed by an analysis of green bonds released by Chinese listed companies between 2015 and 2021, which were usually associated with non-invention patents, especially in firms operating in heavily pollutant sectors, where the public pressure to embrace sustainability is higher (Shi et al. 2023). The absence of clear, actionable standards further exacerbates the issue, impeding meaningful engagement with supply chain actors and partners and making short-term economic concerns outweigh the potential long-term benefits of environmental sustainability, leading to strategic inertia.

This tension between genuine environmental efforts and greenwashing is heightened in cases of aspirational greenwashing, where companies commit to long-term sustainability goals while continuing to carry out unsustainable practices in the short term. Aspirational talk (Christensen et al. 2013), like pledges for carbon neutrality or high-profile CSR goals, can help maintain a company's image and attract environmentally conscious stakeholders. However, it risks creating a gap between promises and what can realistically be achieved, a practice also called "greenwishing." A key example is British Petroleum (BP), which was praised for its sustainability reporting and "Beyond Petroleum" campaign before the 2010 Deepwater Horizon disaster

(Kassinis and Panayiotou 2018). Despite its public commitment to environmental responsibility, BP's operational focus was on cost-cutting and efficiency, including the use of cheaper cementing methods and the reduction of safety barriers—decisions that directly contributed to the catastrophic oil spill (Matejek and Gössling 2014). This revealed a disconnect between its green rhetoric and operational practices. While BP's green image initially boosted its reputation, the disaster underscored the risks of aspirational greenwashing when promises do not align with actions. Moreover, the BP case illustrates the increased scrutiny that companies face when they claim leadership in sustainability. BP's failure to substantiate its "Beyond Petroleum" campaign intensified criticism and resulted in significant reputational damage and a sharp drop in market value due to its loss of moral legitimacy. This underscores the importance of carefully evaluating the strategic benefits and risks of greenwashing as a competitive strategy. While greenwashing may offer short-term competitive advantages, the long-term consequences can be severe. As markets evolve and regulatory pressures increase, companies that rely on greenwashing may face significant reputational risks and financial penalties. This discussion underscores the importance of carefully weighing the strategic benefits and risks of greenwashing.

5.3.3 Consequences of greenwashing

The consequences of greenwashing are multifaceted, affecting both financial and non-financial performance metrics. Companies that engage in greenwashing often experience a range of negative outcomes, from eroded consumer trust to increased scrutiny from regulators and investors. Kahraman and Kazançoğlu (2019) and Nyilasy et al. (2014) demonstrate that greenwashing diminishes consumers' intent to make green purchases as they become more skeptical of environmental claims. This erosion of trust can have long-lasting effects on a company's brand equity and customer loyalty.

From a financial perspective, greenwashing can also have significant repercussions (Bo and Battisti 2024). Cooper and Weber (2021) and García-Sánchez et al. (2021) show how it increases the cost of capital and generally reduces access to finance. Investors are increasingly prioritizing ESG criteria in their decision-making processes, and companies that are perceived as engaging in greenwashing may find it more difficult to secure financing at favorable rates. This is particularly critical in industries where sustainability is a key consideration, such as energy, manufacturing, and consumer goods. A deeper examination reveals that greenwashing negatively impacts overall ESG performance. Indeed, Brandon et al. (2022) and Kim and Yoon (2023) highlight how greenwashing damages a company's reputation and detracts from its ability to meet meaningful sustainability goals. For instance, firms that sign onto voluntary frameworks like the United Nations Principles for Responsible Investment often experience a rise in capital inflows, yet they fail to improve their actual ESG performance. This mismatch between symbolic ESG claims and actual performance can lead to reputational damage and reduced trust from institutional investors (e.g., pension and sovereign wealth funds or foundations), which are increasingly reluctant to back companies with questionable environmental practices as they seek to mitigate long-term financial risks associated with environmental mismanagement. Further elucidating the consequences of greenwashing, Gatti et al. (2021) demon-

strate through an experimental setting that individuals exhibit a diminished inclination to invest in companies implicated in active falsification or engaged in deceptive manipulation. These strategies involve employing tactics that intentionally counterfeit or mislead stakeholders about the company's practices or achievements. This aversion is more pronounced compared to their reaction to companies that practice selective information disclosure, meticulously curating the information released to present themselves in a favorable light. Additionally, the study highlights a comparative tolerance towards companies that adopt a strategy of diverting attention from their shortcomings, prominently showcasing their positive business practices while overshadowing negative misconduct. In a similar vein, investigating US consumers, Ginder et al. (2021) identified different strategic postures that companies may adopt in approaching their CSR communication. They demonstrate that discreet positioning—characterized by companies not openly communicating their CSR actions but still participating in them—is perceived in a similar manner to uniform positioning, where communication and actions are aligned. These findings provide valuable insights into the nuanced perceptions of different greenwashing strategies and CSR communication approaches, further emphasizing the complexity of managing corporate sustainability communications and practices.

Adding to this complexity, Orazalin et al. (2023) studied 592 global firms from 35 countries and found that while higher GHG emissions negatively impact market value, proactive climate change initiatives are positively correlated with market value. However, the presence of board committees on climate change, which are intended to oversee such initiatives, is paradoxically associated with higher GHG emissions. This may suggest that firms with higher emissions feel greater pressure to establish such committees, possibly as a response to stakeholder expectations. Nevertheless, these committees often fail to effectively reduce emissions, serving more as tools for impression management or greenwashing. The findings highlight the need for corporate boards to not only recognize the detrimental effects of carbon emissions on society and the environment but also to integrate genuine, effective climate change initiatives into their business operations. Accordingly, greenwashing can have profound long-term consequences for regulatory relations and legal risks. As Delmas and Burbano (2011) suggest, companies that rely on deceptive environmental claims are more likely to face penalties from regulators as transparency standards evolve and oversight increases. In sectors with stringent environmental regulations, greenwashing can provoke legal challenges or fines that exacerbate financial instability. This dynamic also extends to voluntary reporting frameworks like ISO 14001 or the Global Reporting Initiative, where firms that fail to align their actual environmental practices with reported claims may face reputational backlash, even if they are not legally obligated to meet the same standards.

The complexity of managing greenwashing risks is further compounded by the growing role of stakeholder activism and social media scrutiny. Gatti et al. (2021) and Li et al. (2023a) emphasize how these platforms have empowered consumers and NGOs to hold companies accountable for exaggerated environmental claims. The rapid dissemination of information on social media can intensify public pressure, making it increasingly difficult for companies to control narratives once allegations of greenwashing arise. For instance, in the Volkswagen emissions scandal, the com-

pany's deceptive practices were exposed through social media, leading to significant reputational damage and heightened regulatory scrutiny.

In conclusion, the consequences of greenwashing extend far beyond temporary reputational risks. While some companies may achieve short-term financial gains by enhancing their green image, the long-term costs—such as increased scrutiny, financial penalties, diminished consumer trust, and potential regulatory backlash—can outweigh these benefits. The interconnectedness of financial, reputational, and regulatory outcomes underscores the importance of aligning symbolic environmental communications with substantive actions to avoid the pitfalls of greenwashing.

5.4 Yellow cluster: marketing, perception, and trust

The yellow cluster focuses on the marketing and consumer perception aspects of greenwashing and corporate sustainability claims. Key terms such as “brand trust,” “marketing,” and “conspicuous consumption” highlight the emphasis on consumer–company interactions in the sustainability context. Hence, this cluster emphasizes the psychological and behavioral dimensions of greenwashing, investigating how marketing strategies affect individual consumer skepticism, trust, and brand perceptions in the context of corporate environmental claims. It highlights the complex dynamics between corporate communication strategies and individual consumer responses, underlining the importance of understanding sustainability issues from the consumer's perspective.

5.4.1 Conceptual foundations

This cluster's focus on marketing, consumer perception, and trust in relation to greenwashing is underpinned by several interconnected theoretical frameworks. At the core of this theoretical landscape is signaling theory, which explains how companies attempt to convey their environmental credentials to consumers through various marketing strategies. This theory has been crucial in exploring how firms use green marketing to differentiate themselves and how consumers interpret these signals in the context of potential greenwashing (Berrone et al. 2017). Similarly, the economic aspects of green marketing are addressed through value-based pricing theory, which offers insights into how companies might set prices for sustainable products. This theory helps explain how pricing strategies can influence consumer perceptions of value and quality in relation to green claims, potentially affecting their purchasing decisions. In fact, Lee et al. (2018b) note, price often serves as an external cue for consumers to infer the quality and value of green products.

Closely related to signaling theory is the concept of brand trust, which emerges as a critical factor in the effectiveness of green marketing strategies. Studies have examined how trust mediates the relationship between green marketing efforts and consumer responses, highlighting the challenges companies face in building credibility for their sustainability claims (Chen and Chang 2013). This focus on trust naturally leads to consideration of the theory of planned behavior, which extends our understanding of consumer decision making in sustainability contexts. By suggesting that intentions to engage in environmentally friendly behaviors are influenced by atti-

tudes, subjective norms, and perceived behavioral control, this theory helps explain the often-observed gap between consumers' positive attitudes towards sustainable products and their actual purchasing behaviors.

The cognitive and emotional aspects of consumer responses to sustainability claims are further elucidated through several complementary models. The cognition–affect–behavior (C–A–B) model has been applied to understand how consumers process and respond to sustainability-related information. For instance, Rahman et al. (2015) demonstrated how perceived insincerity in a hotel's green marketing efforts can trigger skepticism (affect), potentially leading to negative behavioral outcomes like reluctance to revisit or support the brand (behavior). Building on this understanding of cognitive and emotional processes, the affect–reason–involvement (ARI) model has also been utilized to explore both logical and emotional elements in green advertising. Schmuck et al. (2018) applied this model to examine how greenwashing advertisements influence consumer evaluations of ads and brands (affect), revealing that emotional responses to nature imagery in ads (reason) can be more influential in shaping attitudes than rational assessments of greenwashing (involvement). Complementing these models, the stimulus-organism-response theory provides a framework for understanding how environmental stimuli (such as green marketing messages) are processed internally by consumers and lead to specific behavioral responses. This theory offers additional insights into the process of consumer decision making in response to sustainability claims. For instance, it could explain how a consumer might respond to a product labeled as “eco-friendly” by researching the claim (organism) before deciding to purchase or not (response). In addition, social exchange theory can be applied to this context to help explain how consumers weigh the perceived benefits and costs of engaging with brands making sustainability claims. This theoretical perspective sheds light on the factors influencing consumer trust and loyalty in the face of green marketing efforts. For example, it can explain why consumers might remain loyal to a brand accused of greenwashing if they perceive the overall benefits of the brand's products to outweigh the costs of the misleading claims.

These theoretical applications collectively illuminate the complexity of consumer responses to green marketing. They highlight that environmental knowledge alone does not guarantee consumers' ability to detect misleading advertising intentions. Instead, the interplay of cognitive, emotional, and economic factors significantly influences how consumers interpret and respond to sustainability claims. By integrating these diverse theories and their applications, researchers have developed a more nuanced understanding of the challenges and dynamics involved in effectively communicating sustainability efforts while maintaining consumer trust and avoiding perceptions of greenwashing. This comprehensive theoretical framework provides a solid foundation for analyzing the multifaceted nature of consumer responses to sustainability marketing and the potential for greenwashing.

5.4.2 Consumer perception and response to greenwashing

Consumer perception and response to greenwashing are complex phenomena influenced by various factors. Research has shown that consumers' ability to detect and respond to greenwashing is not solely dependent on their environmental knowledge.

Instead, as previously introduced, it involves an interplay of cognitive, emotional, and behavioral factors. For instance, Gosselt et al. (2019) found that consumer responses to green initiatives can significantly impact behavior and brand loyalty. However, the effectiveness of these initiatives is mediated by consumers' ability to discern genuine sustainability efforts from greenwashing. This discernment is often challenging, as Chen and Chang (2013) highlight, noting that misleading information can lead to confusion and uncertainty and thus complicate consumers' decision-making processes. In some cases, confusion might arise not only from misleading advertising but also from a company's attempt to obscure the environmental impact of its broader operational practices, thereby deepening the cognitive load required for consumers to make informed choices.

The cognitive challenges posed by greenwashing can lead consumers to adopt different strategies. Some may lean towards heuristic evaluations to conserve cognitive resources, prioritizing cognitive ease over thorough analysis (Testa et al. 2020). This heuristic approach often leads consumers to rely on surface-level indicators, such as green-colored packaging or labels with vague environmental claims like "natural" or "eco-friendly," without thoroughly investigating the actual impact of the product. As a result, products that adopt superficial green cues may still benefit from positive consumer perceptions, even if they lack substantive environmental improvements. Others might experience negative emotions such as a sense of betrayal, which can alter their behaviors in ways that inadvertently undermine genuine sustainability efforts (Kahraman and Kazançoğlu 2019). When consumers perceive that they have been deceived, this emotional response can translate into long-term damage to brand trust, making it difficult for companies to rebuild consumer confidence, even if they later adopt genuine environmental practices.

Interestingly, in this vein, the willingness of consumers to pay a premium for eco-friendly products—linked to substantive CSR activities—can potentially deter companies from greenwashing (Ioannou et al. 2023). However, as Nardi (2022) points out, the effectiveness of this premium depends on consumers' ability to monitor environmental impacts, their CSR preferences, and the competitive environment. In highly competitive markets where multiple firms engage in green marketing, consumers may become overwhelmed by the volume of green claims, which could diminish their ability to distinguish between genuine and misleading CSR initiatives. This, in turn, can lower their willingness to pay a premium for eco-friendly products, as they may perceive these claims to be less trustworthy overall. Schmuck et al. (2018) provide further insights into consumer responses to greenwashing. They found that while ambiguous verbal claims do not necessarily heighten perceptions of greenwashing, outright false claims are perceived as deceptive. Moreover, they discovered that greenwashing with nature imagery triggers an emotional response similar to experiencing real nature, which can be more influential in shaping attitudes towards ads and brands than rational greenwashing perceptions. For instance, an advertisement that uses images of pristine landscapes or wildlife can create a strong positive emotional association with the product, even if the environmental claims made by the company are not entirely credible. This emotional appeal often distracts consumers from critically evaluating the veracity of the claims. Furthermore, Rahman et al. (2015) explored the behavioral consequences of consumer perceptions of insin-

cere green marketing efforts, showing that these perceptions can lead to a significant decline in brand loyalty and negative word-of-mouth. Such negative behaviors can spread quickly through social media platforms, amplifying the reputational damage for companies accused of greenwashing. Indeed, in today's digital environment, consumers are more empowered to share their experiences and influence others' perceptions, making it critical for companies to maintain transparency in their sustainability claims to avoid widespread backlash.

These findings underscore the complexity of consumer responses to greenwashing and highlight the need for a nuanced understanding of how consumers process and react to sustainability claims in marketing communications. Companies that engage in green marketing without substantive environmental commitments risk not only short-term damage to their reputation but also long-term consequences for consumer loyalty and market position. In contrast, firms that invest in transparent, verifiable sustainability practices are more likely to build lasting trust with consumers, even in competitive environments.

5.4.3 Marketing strategies and communication

In the current era of heightened environmental consciousness, companies face significant challenges in effectively communicating their sustainability efforts while avoiding accusations of greenwashing. In fact, companies engaged in green marketing face immediate scrutiny and are often presumed guilty of greenwashing unless they can substantiate their claims (Seele and Gatti 2017). This heightened skepticism has led to a potential boomerang effect, where ostensibly positive intentions can backfire, underscoring the importance of legitimacy and signaling theory in green marketing (Berrone et al. 2017).

Marketing strategies in this domain range from subtle green cues to explicit sustainability claims. For instance, some companies superficially label conventional products as "green" by merely incorporating a minimum of eco-friendly features. This strategy allows firms to avoid substantial investments in truly green innovation while still capitalizing on the growing demand for sustainable products (Guo et al. 2018). A clear example of this is Coca-Cola's introduction of Coca-Cola Life, packaged in green cans and sweetened with stevia, which was marketed as a healthier, environmentally friendly alternative. However, critics pointed out that most of the other ingredients remained unchanged, demonstrating a superficial green shift rather than substantive environmental progress. Similarly, H&M has been criticized for its pledge to use 100% recyclable or sustainable materials by 2030. While the fast fashion giant promotes this goal, critics argue that the sheer volume of production, combined with the complexity of its supply chain, undermines the sustainability message. Many consumers and stakeholders have expressed skepticism over whether the company's green branding aligns with its actual environmental impact.

The effectiveness of these strategies can vary significantly across different sectors. For example, Font et al. (2017) highlighted the risks of misleading or deceptive green advertising in the tourism industry, where consumers are especially sensitive to environmental claims. Similarly, Apaolaza et al. (2023) and Policarpo et al. (2023) showed that brands in the fashion industry frequently face accusations of greenwash-

ing due to opaque supply chains and inconsistent sustainability practices. These industry-specific studies underscore that while greenwashing concerns transcend sector boundaries, the manifestation and impact of greenwashing can vary based on industry characteristics. Corporate communication strategies also play a crucial role in shaping public perception. Siano et al. (2017) demonstrated how scandals such as Volkswagen's emissions cheating can have broad repercussions that transcend the industry and spotlight the critical need for transparency and integrity in corporate environmental initiatives more broadly. The Volkswagen emissions scandal, which involved the company's deceptive marketing of "clean diesel" vehicles, led to significant reputational damage and regulatory consequences. This case highlights how greenwashing not only damages a company's image but also invites legal and regulatory scrutiny, with long-term financial impacts.

The challenges in corporate communication are further complicated by the fact that accusations of greenwashing can arise from both environmental and non-environmental factors, including communication quality. Szabo and Webster (2021) emphasized that such accusations can disproportionately denigrate companies, exceeding the actual impact of their practices. Once implicated in greenwashing, a company may find itself disproportionately blamed for environmental harm or climate change, as public perception can amplify the severity of the offense. Restoring consumer trust after a greenwashing scandal can be a difficult and prolonged process. Guo et al. (2018) highlighted that companies must make significant adjustments in their marketing and communication strategies to regain credibility. For instance, after the Volkswagen scandal, the company embarked on an aggressive rebranding campaign, emphasizing its commitment to electric vehicles and stricter environmental standards. However, regaining trust after such a major breach requires more than just improved communication; it demands substantive changes in both corporate practices and stakeholder engagement strategies.

6 Implications and future research directions

6.1 Theoretical implications

In this section, we address the third and fourth research questions of our study: *RQ3. What are the prevalent theoretical perspectives in greenwashing studies, and how do they interrelate?* and *RQ4. How can future studies address existing gaps in greenwashing research?*

The analysis of the 97 papers in our sample reveals a rich and diverse theoretical landscape underpinning greenwashing research. To synthesize these findings, we propose a multi-level framework that distinguishes between macro, meso, and micro perspectives (Fig. 5). At the macro level (in yellow), theories focus on the broad societal, economic, and institutional systems that shape the context in which greenwashing occurs. The meso level (in blue) delves into the organizational and inter-organizational dynamics that drive greenwashing practices. This level explores how firms develop strategies to navigate internal capabilities and external relationships, often balancing symbolic environmental actions with genuine sustainability efforts.

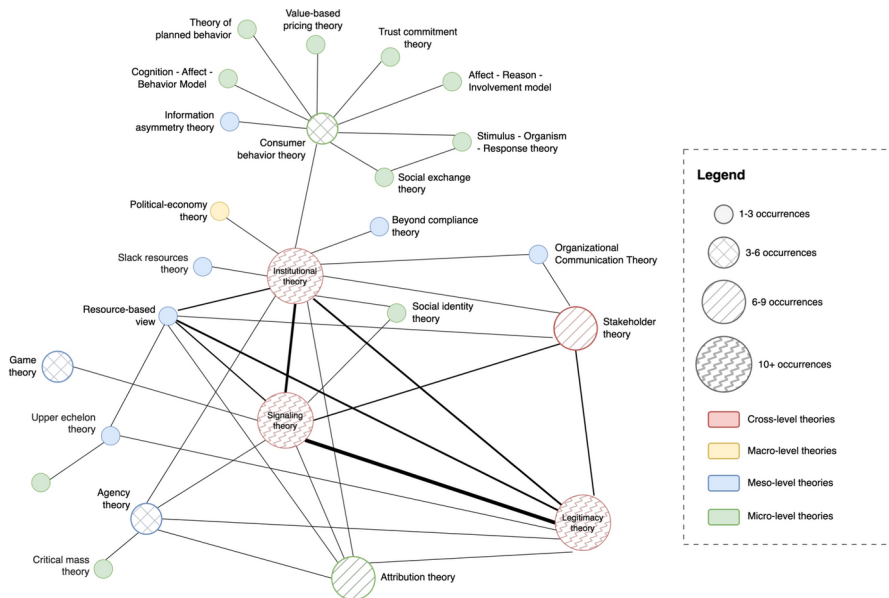


Fig. 5 The theoretical landscape of studies on greenwashing (colour figure online). Source: Own elaboration

At the micro level (in green), theories center on individual behaviors, perceptions, and decision-making processes concerning greenwashing. Additionally, four theories (i.e., institutional, legitimacy, signaling, and stakeholder theory) are applicable across multiple levels, reflecting the complexity of greenwashing as a multi-dimensional phenomenon. These cross-level theories (in red) emphasize how greenwashing practices can simultaneously address societal, organizational, and individual pressures, offering a more integrated view of the phenomenon. For example, stakeholder theory applies at the macro level by examining societal expectations, at the meso level by exploring organizational responses to diverse stakeholder pressures, and at the micro level by analyzing how firms tailor their green claims to individual stakeholders, such as consumers or investors.

To provide a clearer overview of the prominence of each theory and how they interact within the literature, Fig. 5 illustrates the frequency with which these theories appear in the sample through the size of the bubbles, highlighting key interrelationships. Conversely, the thickness of the connecting lines suggests areas where theories cross-fertilize, pointing to potential synergies or areas of underdevelopment. This multi-level approach offers a holistic view of greenwashing, revealing the interplay between societal pressures, organizational strategies, and individual responses. To complement Fig. 5, Table 3 in the Appendix provides a detailed breakdown of each theory, offering definitions and contextualizing their application within greenwashing research. In the following section, we explore the implications of these findings and propose avenues for future research based on the identified gaps.

6.1.1 Macro-level theories

At the macro level, legitimacy theory, institutional theory, and stakeholder theory remain the primary frameworks for understanding how organizations manage greenwashing. These theories explore how firms respond to external pressures such as regulatory changes, cultural norms, and market expectations in order to maintain their legitimacy and ensure their survival. For instance, Marquis et al. (2016) used legitimacy theory to demonstrate how firms selectively disclose environmental information to maintain social approval. However, much of the literature has focused on pragmatic legitimacy, which emphasizes meeting immediate stakeholder needs. Torelli et al. (2020) highlight the importance of legitimacy theory and signaling theory for understanding greenwashing, underscoring the strategic use of selective disclosures to bolster corporate reputation, particularly in industries vulnerable to environmental scrutiny. Accordingly, future research should further investigate how greenwashing impacts the moral and cognitive dimensions of legitimacy. These dimensions extend beyond short-term stakeholder approval and delve into the ethical congruence between a company's stated values and its actions. Research could explore questions such as how greenwashing erodes trust in corporate environmental claims and whether repeated exposure to greenwashing diminishes the moral credibility of firms.

Stakeholder theory also plays a key role at the macro level by examining how firms manage competing stakeholder pressures, often leading to greenwashing as a way to balance diverse expectations. Specifically, instrumental stakeholder theory (Donaldson and Preston 1995) suggests that firms respond to stakeholders not solely out of ethical duty but also to enhance their financial performance. Additionally, stakeholder salience theory (Mitchell et al. 1997) highlights how organizations prioritize responses to stakeholders based on their power, legitimacy, and urgency. This may drive firms to engage in symbolic environmental actions that satisfy the most salient stakeholders while neglecting broader sustainability goals. Despite their relevance, these aspects of stakeholder theory remain underexplored in the greenwashing literature, representing a fertile ground for future research (Kim and Lyon 2015). Accordingly, future studies could investigate how stakeholder salience influences greenwashing strategies—in particular, why some stakeholder pressures are more likely to prompt substantive environmental actions while others trigger symbolic compliance. For example, companies may engage in greenwashing to respond to high-salience stakeholders, such as powerful investors or governmental regulators, who can impact a firm's financial performance or operations. Conversely, firms may downplay concerns from low-salience stakeholders like environmental activists if they perceive the immediate risk of non-compliance as low. Future research could also explore how the effectiveness of green signals varies based on the power and urgency of different stakeholder groups and whether misleading signals result in long-term reputational damage once exposed. Hence, future research could explore how signaling theory intersects with stakeholder salience by investigating how firms use these signals to prioritize their interactions with different stakeholders.

Furthermore, political-economy theory, which examines the role of power structures and political interests in shaping economic systems, has been underutilized in

greenwashing research. This theory offers a valuable lens for exploring how economic and political forces enable firms to engage in greenwashing with minimal consequences. Berrone et al. (2017) argued that weak regulatory frameworks and institutional environments often enable corporate misconduct, particularly in emerging economies. Future studies could investigate how political lobbying and regulatory capture by powerful industries create environments where greenwashing thrives. For example, companies may invest heavily in lobbying efforts to reduce environmental regulations or shape public policy in ways that decrease the risk of accountability for deceptive green claims. One potential avenue for research involves examining the intersection of political-economy theory and institutional theory. This integration could offer insights into how institutional norms and power structures conflate to influence the development of environmental regulations. As discussed, several studies have shown that greenwashing can be seen as a decoupling strategy where firms symbolically comply with environmental regulations while evading substantive change. By combining political-economy and institutional perspectives, researchers could explore how power dynamics shape the enforcement of environmental standards and why certain firms are more successful at avoiding accountability for their greenwashing practices.

Recent developments in emerging technologies further complicate the macro-level dynamics of greenwashing. As Seele and Schultz (2022) describe, the rise of “machinewashing” involves parallels between greenwashing and the unethical use of technologies like AI to manipulate information, conceal environmental wrongdoings, or circumvent regulations. Similarly, Schwaeye et al. (2025) have recently proposed an integrative model for AI adoption, emphasizing the need for robust data management, system integration, and continuous performance monitoring to mitigate risks of misrepresentation and alignment with firms’ sustainability objectives. This represents a new frontier for both greenwashing research and corporate accountability. Future studies could investigate how advanced technologies are leveraged to enhance deceptive green claims, leading to new forms of greenwashing that are more difficult for regulators and stakeholders to detect. Such technological innovations raise questions about the adequacy of current regulatory frameworks for addressing tech-enabled greenwashing.

6.1.2 Meso-level theories

At the meso level, theories such as agency theory, the RBV, and game theory provide valuable insights into how firms balance internal dynamics with external pressures in their environmental strategies. These theories explore the decision-making processes within firms and how firms manage resources to maintain legitimacy and project an environmentally responsible image. In this section, we outline how these theories contribute to understanding greenwashing and propose future research directions.

Agency theory highlights the conflicts of interest that can arise between a firm’s principals (e.g., shareholders) and agents (e.g., managers), particularly when agents prioritize short-term financial gains over long-term sustainability (Gull et al. 2023). Greenwashing can occur when managers mislead stakeholders in order to appear environmentally responsible without making substantive changes to the firm’s prac-

tices. While agency theory has been applied to understand corporate governance issues, future research could explore how incentive structures within firms drive managers toward greenwashing as a short-term solution, especially when sustainability performance metrics are linked to executive compensation. This could be particularly relevant for firms in highly competitive markets where immediate financial performance is crucial. Moreover, this theory aligns well with upper echelon theory, which posits that organizational outcomes are shaped by the values and experiences of top executives (Zhang et al. 2023). For instance, although no studies have integrated these perspectives, leadership characteristics such as risk tolerance or ethical orientation could influence a firm's propensity to engage in symbolic environmental actions. Future research could investigate how these leadership traits mediate the relationship between internal managerial decisions and external greenwashing behaviors, particularly in industries with strong financial pressures.

The RBV and slack resources theory explain how firms use their resources to manage sustainability efforts. The RBV posits that firms gain competitive advantage through the strategic deployment of unique resources and capabilities (Barney 1991). In the context of greenwashing, firms with strong environmental capabilities may genuinely invest in sustainability initiatives, while those lacking such capabilities may resort to symbolic actions to maintain their image. However, slack resources theory offers a more nuanced perspective by suggesting that firms with surplus resources are more likely to innovate and engage in genuine sustainability efforts, while those with fewer resources may adopt symbolic environmental policies to meet stakeholder expectations without making substantive changes (Bryant et al. 2020). This dynamic may be especially pronounced in small and medium-sized enterprises (SMEs), where resource constraints can drive firms toward greenwashing as a cost-effective strategy to manage external pressures. Combining the RBV with slack resources theory could provide a broader framework for understanding how firms' resource availability affects their choice between genuine sustainability efforts and symbolic actions. Accordingly, future research could investigate the role of slack resources in driving meaningful environmental actions, focusing on whether firms with excess resources are better positioned to engage in long-term sustainability strategies. Additionally, studies could explore how resource-constrained firms such as SMEs navigate the tension between stakeholder expectations and resource limitations, leading to either substantive environmental efforts or greenwashing (Orazalin et al. 2023).

Game theory offers another perspective on the strategic decision-making processes related to greenwashing, especially in competitive environments. This theory examines how firms interact with competitors and stakeholders, often using green claims as a strategic tool to differentiate themselves in the market (Huang et al. 2020; Wu et al. 2020). For example, firms may engage in greenwashing to outcompete rivals by projecting an environmentally responsible image while continuing harmful practices behind the scenes. However, the strategic value of greenwashing diminishes as transparency increases, encouraging firms to adopt genuine sustainability measures (Wu et al. 2020). Future research could explore how transparency regimes or consumer access to environmental information influence the effectiveness of greenwashing as a competitive strategy, particularly in emerging markets where regulatory frameworks may be weaker. Several studies have further explored these issues by applying signal-

ing theory to dissect the communicative aspects of greenwashing, where companies often broadcast misleading environmental signals to stakeholders through their CSR disclosures (Kim and Lyon 2015; Torelli et al. 2020) or social media communications (Amin et al. 2022; Lyon and Montgomery 2013). A misalignment here often leads to accusations of greenwashing and corporate hypocrisy, potentially compromising stakeholders' engagement and trust (Bernard and Nicolau 2022; Seele and Gatti 2017). The tension between symbolic management (aimed at creating positive perceptions) and attribution theory (explaining how these symbols are interpreted) highlights the challenges organizations face in managing their environmental image. However, the interplays between what is signaled by companies and what is substantiated by receivers still need deeper investigation (Li et al. 2023a).

Further considering the internal and strategic pressures that influence greenwashing behavior, beyond-compliance theory provides insights into why firms voluntarily exceed regulatory standards in their environmental practices (Bryant et al. 2020). Indeed, while some companies genuinely aim to improve their sustainability performance, others may use beyond-compliance actions as a way to offset past environmental wrongdoings or to create a buffer against future risks. Future research could explore how beyond-compliance strategies are used as defensive mechanisms to mitigate reputational risks associated with greenwashing and how regulatory capture influences the effectiveness of these strategies.

Finally, another underexplored theory is organizational communication theory. Linked to institutional theory and stakeholder theory, organizational communication theory offers perspectives on how firms craft narratives around their sustainability efforts to shape stakeholder perceptions, usually as a response to institutional pressures or regulations. According to Nass and Purwin (2024), organizational narratives and individual sensemaking are key elements in driving sustainability-related decision making. Positioned within the organizational hierarchy, they are actively involved in interpreting (sensemaking) the organizational goals set by senior management and translating (sensegiving) them into actionable steps for employees who carry out day-to-day operations. Future research could explore the roles of other figures, such as lower-level employees, CSR officers, or even external consultants, in influencing sustainability narratives within organizations. Furthermore, research exploring how different layers of organizational communication interact to either support or challenge greenwashing practices could help uncover how various actors contribute to symbolic versus substantive sustainability efforts.

6.1.3 Micro-level theories

At the micro level, theories such as attribution theory, consumer behavior theory, and the theory of planned behavior provide valuable insights into how consumers, employees, and individual stakeholders interpret and react to environmental claims. For instance, Gosselt et al. (2019) used attribution theory to explore how consumers attribute motives to firms' environmental claims, which often leads to skepticism that may persist long-term. While attribution theory has been used to examine how deceptive environmental practices influence consumer behavior, there are ample opportunities to investigate how personal values and individual decision-making pro-

cesses play a role in moderating these effects. Future studies could explore whether more transparent communication can mitigate the negative attributions associated with greenwashing.

Furthermore, consumer behavior theory sheds light on the purchasing decisions made in response to green claims, with a significant impact on consumer trust and loyalty. While much of the existing research focuses on confusion and mistrust (e.g., Apaolaza et al. 2023; Kahraman and Kazançoğlu 2019), there is still a need to explore how consumer education on environmental issues might reduce the effectiveness of greenwashing or the necessity of regulating it (Lee et al. 2018a). Future studies could analyze the relationship between consumer knowledge, transparency of green claims, and purchasing behavior across diverse demographic groups, including younger generations who are increasingly eco-conscious. Moreover, exploring how digital platforms like social media influence these behaviors could open new avenues for research into how consumers interpret and spread environmental messages.

Another emerging area involves stimulus-organism-response (SOR) theory, which provides a more holistic view of how consumers process green marketing stimuli. Rahman and Nguyen-Viet (2023) suggest that trust and non-deception play a key role in how consumers develop trust toward green brands. By using the SOR framework, future research could investigate how green marketing acts as a stimulus that shapes cognitive and emotional responses, potentially leading to changes in behavior or disillusionment when deceptive claims are revealed. These studies could focus on whether specific stimuli (e.g., green certifications and product labels) lead to heightened skepticism or, conversely, reinforce trust in genuine environmental initiatives.

Gender socialization theory and critical mass theory offer additional underexplored avenues for examining how individual characteristics and group dynamics influence perceptions of greenwashing. For instance, gender socialization theory suggests that men and women may process environmental claims differently due to varied social conditioning regarding environmental responsibility. Future research could investigate how gendered perceptions influence responses to green claims, especially in light of growing interest in gender diversity in corporate leadership (Sterbenk et al. 2022). Additionally, critical mass theory, which has been applied to investigate organizational board composition (Eliwa et al. 2023), can be extended to the consumer level to explore whether a critical mass of informed consumers can shift market dynamics and reduce the prevalence of greenwashing. Research could examine how social media platforms and online consumer communities help to generate this critical mass and influence broader purchasing behaviors.

Another key area of interest is information asymmetry theory; although it is traditionally applied in agency contexts, it has significant relevance for the micro level. Greenwashing exploits the information gap between companies and consumers, particularly by making misleading environmental claims that are difficult for the average consumer to verify. Future studies could examine how reducing information asymmetry affects consumer and investor trust and purchasing behavior or investment decisions (Gatti et al. 2021; Koseoglu et al. 2021). This line of research could extend to investigating how digital tools like blockchain-based transparency initiatives might reduce information asymmetry and protect stakeholders from deceptive practices.

In conclusion, micro-level theories offer substantial opportunities for future research, especially for understanding consumer behavior and individual responses to greenwashing. As the cross-level theoretical integrations in this framework suggest, combining macro-level theories such as institutional theory with micro-level approaches like consumer behavior theory could provide a more comprehensive understanding of how regulatory pressures shape individual decision making. Similarly, integrating meso-level organizational communication theory with micro-level theories could reveal how firms' communication strategies influence individual trust and behavior. In this vein, future studies could further explore how group dynamics, individual characteristics, and information transparency affect the perceived legitimacy of environmental claims and, ultimately, how these perceptions influence broader market behaviors.

6.2 Practical implications

Beyond theoretical contributions, this study offers valuable insights for practitioners as well. For businesses, the findings underscore the critical importance of aligning green claims with substantive environmental actions. This alignment not only mitigates the risks of reputational damage and loss of stakeholder trust but also strengthens accountability within organizations. For businesses, the findings also highlight the importance of aligning environmental claims with substantive sustainability actions, as greenwashing poses significant risks to corporate reputation, stakeholder trust, and long-term success. Companies that engage in symbolic or misleading environmental efforts face the potential for reputational damage, especially in an era when consumers, investors, and regulators increasingly demand transparency. The multi-level framework presented in this study can serve as a tool for firms to evaluate how greenwashing impacts various aspects of their operations, from corporate strategy to consumer behavior and stakeholder engagement. By understanding these dynamics, businesses can better navigate the complexities of sustainable branding, ensuring that their claims are backed by authentic environmental practices. For policymakers, the study emphasizes the need for stronger regulatory frameworks that promote transparency and accountability in green marketing. The cross-level theoretical insights offer guidance on how to detect and prevent greenwashing, supporting the development of policies that not only penalize deceptive practices but also encourage firms to make meaningful environmental contributions. In this way, the research contributes to both the academic discourse and the practical efforts of businesses and regulators to foster genuine sustainability in an increasingly scrutinized market.

6.3 Concluding remarks

The evolution of greenwashing research underscores its growing relevance across a diverse range of sectors, from large corporations to SMEs, public organizations, and online communities. As businesses and policymakers increasingly confront the complexities of environmental integrity in a rapidly evolving sustainable and digital society, the importance of understanding greenwashing continues to rise. This review offers significant insights into how greenwashing is framed, communicated, and per-

ceived, and it provides valuable guidance for both researchers and practitioners. The bibliometric analysis sheds light on the key trends in greenwashing research (RQ1), identifying the leading publications, journals, and contributors that have shaped the field. By outlining the most influential works and authors, this analysis maps the intellectual landscape of greenwashing studies and thus allows researchers and practitioners alike to identify pivotal contributions and track the growth of scholarly interest in the topic. The thematic discussion based on distinct clusters (RQ2) illustrates how various theoretical perspectives have informed the conceptualization of greenwashing. These clusters reveal how greenwashing is not only a business practice but also an ethical and communicative challenge with significant implications for corporate reputation and stakeholder engagement. By grouping the literature into themes such as corporate communication, regulatory compliance, and stakeholder influence, we highlight the diverse contexts in which greenwashing is studied. This thematic breakdown also enables businesses to understand the multiple dimensions of greenwashing, from the ethical dilemmas it presents to the practical strategies for managing stakeholder perceptions. The multi-level theoretical framework distinguishes between macro-, meso-, and micro-level theories while emphasizing the cross-level nature of many of them (RQ3). Institutional theory, signaling theory, stakeholder theory, and legitimacy theory are especially relevant across all levels. At the macro level, these theories explore how external pressures, including regulatory norms and societal expectations, influence corporate greenwashing practices. At the meso level, they examine how organizations manage legitimacy and stakeholder expectations, balancing internal strategies with external demands. At the micro level, these theories analyze how individuals within organizations, such as employees or managers, perceive and respond to green claims, shaping internal decision making. Recognizing the cross-level relevance of these theories allows for a more integrated understanding of greenwashing, encouraging research that bridges these different analytical layers. Finally, we leveraged the theoretical framework to identify key gaps and future research directions (RQ4).

However, like any study, this research has limitations that open up avenues for further development. First, the reliance on the Scopus and Web of Science databases may have excluded relevant gray literature or contributions from other fields. Second, while we focused on the business and management literature, future studies could incorporate perspectives from adjacent disciplines to enrich the theoretical landscape. Third, the thematic analysis of the literature may be subject to interpretation, particularly when identifying and categorizing clusters. While the methodology is robust, variations in interpretation are possible. Future research could benefit from further refining the methodology for cluster identification, potentially incorporating alternative approaches or validation techniques to ensure a comprehensive representation of themes within greenwashing studies. By addressing these limitations, future research can continue to develop a more integrated and multidisciplinary approach to greenwashing, contributing to both theoretical advancement and practical solutions. The identification of key gaps and the proposal of cross-level theoretical integration provide a roadmap for future studies to deepen the understanding of greenwashing dynamics. This review underscores the need for continued exploration of greenwashing across multiple levels of analysis, which can ultimately inform both academic

discourse and corporate practice. As environmental concerns grow increasingly central to organizational strategies, addressing these gaps will be crucial to foster more transparent, accountable, and sustainable business practices.

Appendix

See Table 3.

Table 3 List of theories applied in greenwashing research, categorized by levels and ordered by frequency of occurrence (Freq.).

Source: Own elaboration

Theory	Freq	Definition	Application to greenwashing
<i>Cross-level theories</i>			
Institutional theory	15	Organizations are influenced by institutional norms, rules, and pressures	Macro-level applications: it analyzes how regulatory environments and industry norms shape firms' greenwashing practices and their responses to institutional pressures Meso-level applications: it investigates how institutional norms impact organizational strategies for environmental claims Micro-level applications: it explores how individuals within firms internalize institutional pressures, leading to actions that either align with or counteract greenwashing
Signaling theory	15	Organizations send signals to the market through their actions or communications	Macro-level applications: it analyzes how firms use green claims as signals of environmental commitment, often without substantive actions to back them up Meso-level applications: it investigates how firms shape their green marketing messages as signals of performance and commitment Micro-level applications: it looks at how consumers or investors interpret signals sent by organizations and assess their credibility based on perceived honesty and past actions
Legitimacy theory	12	This theory explores how organizations seek to justify their existence and actions to external stakeholders to gain legitimacy	Macro-level applications: it analyzes how greenwashing helps firms gain legitimacy by creating a false impression of environmental responsibility, thus maintaining their social license to operate Meso-level applications: it investigates how firms act to maintain legitimacy with key stakeholders, often without substantive changes Micro-level applications: it examines how individuals or groups within the organization may justify or challenge deceptive practices aimed at maintaining legitimacy

Table 3 (continued)

Theory	Freq	Definition	Application to greenwashing
Stakeholder theory	8	Firms must balance the interests of various stakeholders	Macro-level applications: it examines how firms manage competing stakeholder pressures, often leading to greenwashing when trying to satisfy diverse environmental expectations Meso-level applications: it focuses on organizational efforts to communicate environmental responsibility to different stakeholder groups Micro-level applications: it explores how individual stakeholders (e.g., consumers, employees) perceive and respond to greenwashing practices
<i>Macro-level theories</i>			
Political-economy theory	1	The economic system is shaped by power structures and political interests	It explores how economic and political influences allow firms to engage in greenwashing to maintain their market position under perceived environmental pressures
<i>Meso-level theories</i>			
Agency theory	4	This theory examines conflicts of interest between principals and agents	It discusses how greenwashing may arise when managers prioritize short-term financial gains over long-term sustainability, misleading shareholders
Game theory	4	This theory analyzes strategic interactions where the outcome for each player depends on the actions of others	It explores competitive strategies in which firms may engage in greenwashing to outdo competitors in the sustainability race without substantive actions
Organizational communication theory	3	This theory focuses on how organizations communicate internally and externally	It investigates how firms structure their communication strategies around environmental claims, often engaging in greenwashing to shape public perception
Resource-based view (RBV)	3	Firms gain competitive advantage through unique resources and capabilities	It discusses how firms use their environmental capabilities or lack thereof to justify environmental claims, often leading to greenwashing
Beyond-compliance theory	1	This theory suggests that firms voluntarily exceed regulatory standards to gain competitive advantages	Companies might go beyond mere compliance in environmental claims but still engage in greenwashing by exaggerating their efforts
Slack resources theory	1	Firms with excess resources are more likely to innovate and outperform	It explains how companies with more slack resources may invest in substantive CSR initiatives, while resource-scarce firms are more prone to symbolic actions and greenwashing
Social exchange theory	1	Social behavior is the result of an exchange process aiming to maximize benefits and minimize costs	It studies how firms engage in greenwashing to gain trust and support from stakeholders by projecting a false environmental image
<i>Micro-level theories</i>			
Attribution theory	7	This theory explores how individuals attribute causes to events or behaviors	It focuses on how consumers interpret and respond to greenwashing, attributing deceptive motives to firms' environmental claims

Table 3 (continued)

Theory	Freq	Definition	Application to greenwashing
Consumer behavior theory	5	This theory studies how individuals make decisions about products and services	It investigates how greenwashing influences consumer choices, often leading to confusion and mistrust of genuine environmental efforts
Social identity theory	2	This theory focuses on how individuals define themselves in relation to groups and how group identity affects behavior	Greenwashing can manipulate social identity by appealing to consumers' desire to belong to a group that values sustainability, even if the claims are false
Stimulus-organism-response theory	2	This theory explains how environmental stimuli influence individual responses through cognitive and emotional processes	It analyzes how green marketing acts as a stimulus that shapes consumer perceptions and behaviors, sometimes leading to disillusionment when claims are found to be false
Cognition-affect-behavior model	1	This theory explores how cognitive processes, emotions, and behaviors are interrelated in decision making	It examines how consumers' cognitive and emotional responses are manipulated by firms' misleading environmental claims, impacting their behavior
Critical mass theory	1	This theory states that a certain number of individuals within a group is necessary to effect change	In the context of greenwashing, this theory is used to examine how a critical mass of women or independent directors on boards can push for substantive environmental actions
Gender socialization theory	1	This theory suggests that social behaviors and roles are learned based on gender expectations	It is applied to study the impact of having a critical mass of women or independent directors in organizations, influencing the choice to take substantive sustainability actions over greenwashing
Theory of planned Behavior	1	This theory predicts individual behavior based on attitudes, subjective norms, and perceived behavioral control	It analyzes how consumers' attitudes and perceptions about green claims influence their responses to greenwashed products or firms
Trust commitment theory	1	This theory focuses on the role of trust and commitment in relationships	It investigates how greenwashing damages consumer trust, affecting their long-term commitment to brands or firms engaging in deceptive practices
Value-based pricing theory	1	This theory suggests that prices should reflect the perceived value to the customer	It discusses how firms may use greenwashing to justify premium pricing, despite offering little environmental value in their products

Acknowledgements This study was funded by the European Union—NextGenerationEU, Mission 4, Component 2, in the framework of the GRINS -Growing Resilient, Inclusive and Sustainable project (GRINS PE00000018– CUP D13C22002160001). The views and opinions expressed are solely those of the authors and do not necessarily reflect those of the European Union, nor can the European Union be held responsible for them.

Funding Open access funding provided by Università degli Studi di Torino within the CRUI-CARE Agreement.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative

Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- *Ali Gull A, Hussain N, Akbar Khan S, Nadeem M, Mansour Zalata A (2023) Walking the talk? A corporate governance perspective on corporate social responsibility decoupling. *Br J Manag* 34(4):2186–2211. <https://doi.org/10.1111/1467-8551.12695>
- Amin MH, Ali H, Mohamed EKA (2022) Corporate social responsibility disclosure on Twitter: signalling or greenwashing? Evidence from the UK. *Int J Financ Econ*. <https://doi.org/10.1002/ijfe.2762>
- Amores-Salvado J, Martin-de Castro G, Albertini E (2023) Walking the talk, but above all, talking the walk: looking green for market stakeholder engagement. *Corp Soc Responsib Environ Manag* 30(1):431–442. <https://doi.org/10.1002/csr.2364>
- *Apaolaza V, Policarpo MC, Hartmann P, Paredes MR, D'Souza C (2023) Sustainable clothing: why conspicuous consumption and greenwashing matter. *Bus Strateg Environ* 32(6):3766–3782. <https://doi.org/10.1002/bse.3335>
- Aria M, Cuccurullo C (2017) bibliometrix: an R-tool for comprehensive science mapping analysis. *J Informet* 11(4):959–975. <https://doi.org/10.1016/j.joi.2017.08.007>
- Baima G, Forliano C, Santoro G, Vrontis D (2020) Intellectual capital and business model: a systematic literature review to explore their linkages. *J Intellect Cap*. <https://doi.org/10.1108/JIC-02-2020-0055>
- Bandini F, Chiappini H, Pallara F (2022) Fund managers acting as impact investors: strategies, practices, and tensions. *Corp Soc Responsib Environ Manag* 29(4):1084–1095. <https://doi.org/10.1002/csr.2255>
- Barbu EM, Ionescu-Felegă L, Ferrat Y (2022) The evolution of environmental reporting in Europe: the role of financial and non-financial regulation. *Int J Account*. <https://doi.org/10.1142/S1094406022500081>
- Barney J (1991) Firm resources and sustained competitive advantage. *J Manag* 17(1):99–120
- Battisti E, Nirino N, Leonidou E, Thrassou A, Vrontis D (2023) Mergers and acquisitions and environmental conditions: a systematic literature review to investigate their linkage. *Manag Res Rev* 47(2):226–244. <https://doi.org/10.1108/MRR-09-2021-0687>
- *Bernard S, Nicolau JL (2022) Environmental certification and hotel market value. *Int J Hosp Manag*. <https://doi.org/10.1016/j.ijhm.2021.103129>
- Bernini F, La Rosa F (2023) Research in the greenwashing field: concepts, theories, and potential impacts on economic and social value. *J Manag Gov*. <https://doi.org/10.1007/s10997-023-09686-5>
- *Berrone P, Fosfuri A, Gelabert L (2017) Does greenwashing pay off? Understanding the relationship between environmental actions and environmental legitimacy. *J Bus Ethics* 144(2):363–379. <https://doi.org/10.1007/s10551-015-2816-9>
- Bertello A, De Bernardi P, Ricciardi F (2023) Open innovation: status quo and quo vadis-an analysis of a research field. *Rev Manag Sci* 18:633–683. <https://doi.org/10.1007/s11846-023-00655-8>
- Blazkova T, Pedersen ERG, Andersen KR, Rosati F (2023) Greenwashing debates on Twitter: stakeholders and critical topics. *J Clean Prod*. <https://doi.org/10.1016/j.jclepro.2023.139260>
- Bo S, Battisti E (2024) Green finance and greenwashing: charting a sustainable path forward. *Qual Res Financ Mark*. <https://doi.org/10.1108/QRFM-05-2024-0125>
- Bowen F, Aragon-Correa JA (2014) Greenwashing in corporate environmentalism research and practice: the importance of what we say and do. *Organ Environ* 27(2):107–112
- *Brandon RG, Glossner S, Krueger P, Matos P, Steffen T (2022) Do responsible investors invest responsibly? *Rev Financ* 26(6):1389–1432. <https://doi.org/10.1093/rof/rfac064>
- *Bryant A, Griffin JJ, Perry VG (2020) Mitigating climate change: a role for regulations and risk-taking. *Bus Strateg Environ* 29(2):605–618. <https://doi.org/10.1002/bse.2391>

- Buchkremer R, Demund A, Ebener S, Gampfer F, Jägering D, Jürgens A, Klenke S, Krimpmann D, Schmank J, Spiekermann M (2019) The application of artificial intelligence technologies as a substitute for reading and to support and enhance the authoring of scientific review articles. *IEEE Access* 7:65263–65276
- Chartered Association of Business Schools (2021) Academic journal guide 2021. <https://chartereddabs.org/academic-journal-guide/academic-journal-guide-2021>. Accessed 29 Nov 2023
- Chatterjee S, Chaudhuri R, Vrontis D, Bresciani S (2024) Exploring the effect of government incentives on electric vehicle purchase intention in smart cities. *J Clean Prod* 477:143841. <https://doi.org/10.1016/j.jclepro.2024.143841>
- *Chen YS, Chang CH (2013) Greenwash and green trust: the mediation effects of green consumer confusion and green perceived risk. *J Bus Ethics* 114(3):489–500. <https://doi.org/10.1007/s10551-012-1360-0>
- Christensen LT, Morsing M, Thyssen O (2013) CSR as aspirational talk. *Organization* 20(3):372–393. <https://doi.org/10.1177/1350508413478310>
- Christofi M, Leonidou E, Vrontis D (2017) Marketing research on mergers and acquisitions: a systematic review and future directions. *Int Mark Rev* 34(5):629–651
- *Cooper LA, Weber J (2021) Does benefit corporation status matter to investors? An exploratory study of investor perceptions and decisions. *Bus Soc* 60(4):979–1008. <https://doi.org/10.1177/0007650319898462>
- De Bernardi P, Bertello A, Forlano C (2021) Digital platforms for circular business model innovation. In: Riad Shams SM, Vrontis D, Weber Y, Tsoukatos Rogdia E, Santoro G (eds) *Business model innovation*. Routledge, pp 116–131. <https://doi.org/10.4324/9781003028307-8-8>
- De Meyst KJ, Cardinaels E, Van den Abbeele A (2023) CSR disclosures in buyer-seller markets: the impact of assurance of CSR disclosures and incentives for CSR investments. *Account Organ Soc*. <https://doi.org/10.1016/j.aos.2023.101498>
- de Freitas Netto SV, Sobral MFF, Ribeiro ARB, Soares GRDL (2020) Concepts and forms of greenwashing: a systematic review. *Environ Sci Eur* 32(1):1–12. <https://doi.org/10.1186/s12302-020-0300-3>
- *Delmas MA, Burbano VC (2011) The drivers of greenwashing. *Calif Manag Rev* 54(1):64–87. <https://doi.org/10.1525/cmr.2011.54.1.64>
- Deschryver P, de Mariz F (2020) What future for the green bond market? How can policymakers, companies, and investors unlock the potential of the green bond market? *J Risk Financ Manag*. <https://doi.org/10.3390/jrfm13030061>
- Donaldson T, Preston LE (1995) The stakeholder theory of the corporation: concepts, evidence, and implications. *Acad Manag Rev* 20(1):65–91. <https://doi.org/10.5465/amr.1995.9503271992>
- Dorflleitner G, Utz S (2023) Green, green, it's green they say: a conceptual framework for measuring greenwashing on firm level. *Rev Manag Sci*. <https://doi.org/10.1007/s11846-023-00718-w>
- *Eliwa Y, Aboud A, Saleh A (2023) Board gender diversity and ESG decoupling: Does religiosity matter? *Bus Strateg Environ* 32(7):4046–4067. <https://doi.org/10.1002/bse.3353>
- *Fernando AG, Suganthi L, Sivakumaran B (2014) If you blog, will they follow? Using online media to set the agenda for consumer concerns on 'greenwashed' environmental claims. *J Advert* 43(2):167–180. <https://doi.org/10.1080/00913367.2013.841088>
- Ferrón-Vílchez V, Valero-Gil J, Suárez-Perales I (2021) How does greenwashing influence managers' decision-making? An experimental approach under stakeholder view. *Corp Soc Responsib Environ Manag* 28(2):860–880. <https://doi.org/10.1002/csr.2095>
- *Fiechter P, Hitz JM, Lehmann N (2022) Real effects of a widespread CSR reporting mandate: evidence from the European Union's CSR directive. *J Account Res* 60(4):1499–1549. <https://doi.org/10.1111/1475-679X.12424>
- Follmer KB, Jones KS (2018) Mental illness in the workplace: an interdisciplinary review and organizational research agenda. *J Manag* 44(1):325–351
- *Font X, Elgammal I, Lamond I (2017) Greenhushing: the deliberate under communicating of sustainability practices by tourism businesses. *J Sustain Tour* 25(7):1007–1023. <https://doi.org/10.1080/09669582.2016.1158829>
- Forlano C, De Bernardi P, Yahiaoui D (2021) Entrepreneurial universities: a bibliometric analysis within the business and management domains. *Technol Forecast Soc Change*. <https://doi.org/10.1016/j.techfore.2020.120522>
- *García-Sánchez IM, Hussain N, Khan SA, Martínez-Ferrero J (2021) Do markets punish or reward corporate social responsibility decoupling? *Bus Soc* 60(6):1431–1467. <https://doi.org/10.1177/0007650319898839>

- Gatti L, Seele P, Rademacher L (2019) Grey zone in–greenwash out. A review of greenwashing research and implications for the voluntary–mandatory transition of CSR. *Int J Corp Soc Responsib* 4(1):1–15
- *Gatti L, Pizzetti M, Seele P (2021) Green lies and their effect on intention to invest. *J Bus Res* 127:228–240. <https://doi.org/10.1016/j.jbusres.2021.01.028>
- *Gilbert B, James A, Shogren JF (2018) Corporate apology for environmental damage. *J Risk Uncertain* 56(1):51–81. <https://doi.org/10.1007/s11166-018-9276-4>
- *Ginder W, Kwon WS, Byun SE (2021) Effects of internal–external congruence-based CSR positioning: an attribution theory approach. *J Bus Ethics* 169(2):355–369. <https://doi.org/10.1007/s10551-019-04282-w>
- *Gosselt JF, van Rompay T, Haske L (2019) Won't get fooled again: the effects of internal and external CSR ECO-labeling. *J Bus Ethics* 155(2):413–424. <https://doi.org/10.1007/s10551-017-3512-8>
- *Graafland J, Smid H (2019) Decoupling among CSR policies, programs, and impacts: an empirical study. *Bus Soc* 58(2):231–267. <https://doi.org/10.1177/0007650316647951>
- *Gull AA, Hussain N, Khan SA, Khan Z, Saeed A (2023) Governing corporate social responsibility decoupling: the effect of the Governance Committee on Corporate Social Responsibility Decoupling. *J Bus Ethics* 185(2):349–374. <https://doi.org/10.1007/s10551-022-05181-3>
- *Guo R, Tao L, Li CB, Wang T (2017) A path analysis of greenwashing in a trust crisis among Chinese energy companies: the role of brand legitimacy and brand loyalty. *J Bus Ethics* 140(3):523–536. <https://doi.org/10.1007/s10551-015-2672-7>
- *Guo R, Zhang W, Wang T, Li CB, Tao L (2018) Timely or considered? Brand trust repair strategies and mechanism after greenwashing in China—from a legitimacy perspective. *Ind Mark Manag* 72:127–137. <https://doi.org/10.1016/j.indmarman.2018.04.001>
- Hart SL, Dowell G (2011) A natural-resource-based view of the firm: fifteen years after. *J Manag* 37(5):1464–1479. <https://doi.org/10.1177/0149206310390219>
- *Hassan AM, Roberts L, Atkins J (2020) Exploring factors relating to extinction disclosures: What motivates companies to report on biodiversity and species protection? *Bus Strateg Environ* 29(3):1419–1436. <https://doi.org/10.1002/bse.2442>
- *Heras-Saizarbitoria I, Boiral O, Díaz de Junguitu A (2020) Environmental management certification and environmental performance: Greening or greenwashing? *Bus Strateg Environ* 29(6):2829–2841. <https://doi.org/10.1002/bse.2546>
- *Huang H, Xing X, He Y, Gu X (2020) Combating greenwashers in emerging markets: a game-theoretical exploration of firms, customers and government regulations. *Transp Res Part E Log Transp Rev*. <https://doi.org/10.1016/j.tre.2020.101976>
- Inês A, Diniz A, Moreira AC (2023) A review of greenwashing and supply chain management: challenges ahead. *Clean Environ Syst* 11:100136
- *Ioannou I, Kassinis G, Papagiannakis G (2023) The impact of perceived greenwashing on customer satisfaction and the contingent role of capability reputation. *J Bus Ethics* 185(2):333–347. <https://doi.org/10.1007/s10551-022-05151-9>
- *Kahraman A, Kazançoğlu İ (2019) Understanding consumers' purchase intentions toward natural-claimed products: a qualitative research in personal care products. *Bus Strateg Environ* 28(6):1218–1233. <https://doi.org/10.1002/bse.2312>
- *Kassinis G, Panayiotou A (2018) Visuality as greenwashing: the case of BP and Deepwater Horizon. *Organ Environ* 31(1):25–47. <https://doi.org/10.1177/1086026616687014>
- Kassinis GI, Kay AA, Papagiannakis G, Vlachos PA (2022) Stigma as moral insurance: how stigma buffers firms from the market consequences of greenwashing. *J Manag Stud* 59(8):2154–2190. <https://doi.org/10.1111/joms.12873>
- Keupp MM, Palmié M, Gassmann O (2012) The strategic management of innovation: a systematic review and paths for future research. *Int J Manag Rev* 14(4):367–390. <https://doi.org/10.1111/j.1468-2370.2011.00321.x>
- *Kim EH, Lyon TP (2015) Greenwash vs. brownwash: exaggeration and undue modesty in corporate sustainability disclosure. *Organ Sci* 26(3):705–723. <https://doi.org/10.1287/orsc.2014.0949>
- *Kim S, Yoon A (2023) Analyzing active fund managers' commitment to ESG: evidence from the United Nations Principles for Responsible Investment. *Manag Sci* 69(2):741–758. <https://doi.org/10.1287/mnsc.2022.4394>
- *Koseoglu MA, Uyar A, Kilic M, Kuzey C, Karaman AS (2021) Exploring the connections among CSR performance, reporting, and external assurance: evidence from the hospitality and tourism industry. *Int J Hosp Manag*. <https://doi.org/10.1016/j.ijhm.2020.102819>

- Kossyva D, Theriou G, Aggelidis V, Sarigiannidis L (2023) Definitions and antecedents of engagement: a systematic literature review. *Manag Res Rev* 46(5):719–738
- Kraus S, Breier M, Dasí-Rodríguez S (2020) The art of crafting a systematic literature review in entrepreneurship research. *Int Entrep Manag J* 16(3):1023–1042
- Kraus S, Breier M, Lim WM, Dabić M, Kumar S, Kanbach D, Mukherjee D, Corvello V, Piñeiro-Chousa J, Liguori E (2022) Literature reviews as independent studies: guidelines for academic practice. *RMS* 16(8):2577–2595
- *Laufer WS (2003) Social accountability and corporate greenwashing. *J Bus Ethics* 43(3):253–261. <https://doi.org/10.1023/A:1022962719299>
- *Lee MT, Raschke RL (2023) Stakeholder legitimacy in firm greening and financial performance: What about greenwashing temptations? *J Bus Res*. <https://doi.org/10.1016/j.jbusres.2022.113393>
- *Lee HCB, Cruz JM, Shankar R (2018a) Corporate social responsibility (CSR) issues in supply chain competition: Should greenwashing be regulated? *Decis Sci* 49(6):1088–1115. <https://doi.org/10.1111/deci.12307>
- *Lee J, Bhatt S, Suri R (2018b) When consumers penalize not so green products. *Psychol Mark* 35(1):36–46. <https://doi.org/10.1002/mar.21069>
- *Li W, Li W, Seppänen V, Koivumäki T (2023a) Effects of greenwashing on financial performance: moderation through local environmental regulation and media coverage. *Bus Strateg Environ* 32(1):820–841. <https://doi.org/10.1002/bse.3177>
- Li W, Liu Y, Wang L (2023b) Moral mentor of the company? Multifaceted influence of sustainable and responsible funds on corporate social responsibility disclosure in China. *Asia Pac J Manag*. <https://doi.org/10.1007/s10490-023-09900-w>
- Liu M, Luo X, Lu WZ (2023) Public perceptions of environmental, social, and governance (ESG) based on social media data: evidence from China. *J Clean Prod*. <https://doi.org/10.1016/j.jclepro.2022.135840>
- Lyon TP, Maxwell JW (2011) Greenwash: corporate environmental disclosure under threat of audit. *J Econ Manag Strateg* 20(1):3–41. <https://doi.org/10.1111/j.1530-9134.2010.00282.x>
- *Lyon TP, Montgomery AW (2013) Tweetjacked: the impact of social media on corporate greenwash. *J Bus Ethics* 118(4):747–757. <https://doi.org/10.1007/s10551-013-1958-x>
- *Lyon TP, Montgomery AW (2015) The means and end of greenwash. *Organ Environ* 28(2):223–249. <https://doi.org/10.1177/1086026615575332>
- *Mahoney LS, Thorne L, Cecil L, LaGore W (2013) A research note on standalone corporate social responsibility reports: signaling or greenwashing? *Crit Perspect Account* 24(4–5):350–359. <https://doi.org/10.1016/j.cpa.2012.09.008>
- *Marquis C, Toffel MW, Zhou Y (2016) Scrutiny, norms, and selective disclosure: a global study of greenwashing. *Organ Sci* 27(2):483–504. <https://doi.org/10.1287/orsc.2015.1039>
- Marshall D, Rehme J, O'Dochartaigh A, Kelly S, Boojihawon R, Chicksand D (2023) Reporting controversial issues in controversial industries. *Account Audit Account J* 36(9):483–512. <https://doi.org/10.1108/AAAJ-07-2020-4684>
- Martín-Navarro A, Lechuga Sancho MP, Martínez-Fierro S (2023) Evolution of entrepreneurship research in the food sector: a bibliometric review. *Br Food J* 125(5):1537–1558
- *Matejek S, Gössling T (2014) Beyond legitimacy: a case study in BP's 'green lashing.' *J Bus Ethics* 120(4):571–584. <https://doi.org/10.1007/s10551-013-2006-6>
- Merigó JM, Cancino CA, Coronado F, Urbano D (2016) Academic research in innovation: a country analysis. *Scientometrics* 108(2):559–593. <https://doi.org/10.1007/s11192-016-1984-4>
- Mitchell RK, Agle BR, Wood DJ (1997) Toward a theory of stakeholder identification and salience: defining the principle of who and what really counts. *Acad Manag Rev* 22(4):853–886. <https://doi.org/10.5465/AMR.1997.9711022105>
- Montero-Navarro A, González-Torres T, Rodríguez-Sánchez JL, Gallego-Losada R (2021) A bibliometric analysis of greenwashing research: a closer look at agriculture, food industry and food retail. *Br Food J* 123(13):547–560. <https://doi.org/10.1108/BFJ-06-2021-0708>
- Montgomery AW, Lyon TP, Barg J (2023) No end in sight? A greenwash review and research agenda. *Organ Environ*. <https://doi.org/10.1177/10860266231168905>
- Moodaley W, Telukdarie A (2023) Greenwashing, sustainability reporting, and artificial intelligence: a systematic literature review. *Sustainability* 15(2):1481
- Motavalli J (2011) A history of greenwashing: how dirty towels impacted the green movement. *Daily Financ* 12:02–11

- *Munasinghe A, Cuckston T, Rowbottom N (2021) Sustainability certification as marketisation: rainforest alliance in the Sri Lankan tea production industry. *Account Forum* 45(3):247–272. <https://doi.org/10.1080/01559982.2021.1893053>
- *Nardi L (2022) The corporate social responsibility price premium as an enabler of substantive CSR. *Acad Manag Rev* 47(2):282–308. <https://doi.org/10.5465/amr.2019.0425>
- Nass L, Purwin N (2024) From words to action: unraveling the role of organizational. *Acad Manag J* 38(4):997–1035
- *Nyilasy G, Gangadharbatla H, Paladino A (2014) Perceived greenwashing: the interactive effects of green advertising and corporate environmental performance on consumer reactions. *J Bus Ethics* 125(4):693–707. <https://doi.org/10.1007/s10551-013-1944-3>
- Orazalin NS, Ntim CG, Malagila JK (2023) Board sustainability committees, climate change initiatives, carbon performance, and market value. *Br J Manag*. <https://doi.org/10.1111/1467-8551.12715>
- Öztürk O, Kocaman R, Kanbach DK (2024) How to design bibliometric research: an overview and a framework proposal. *RMS* 18:333–3361. <https://doi.org/10.1007/s11846-024-00738-0>
- *Parguel B, Benoît-Moreau F, Larceux F (2011) How sustainability ratings might deter ‘greenwashing’: a closer look at ethical corporate communication. *J Bus Ethics* 102(1):15–28. <https://doi.org/10.1007/s10551-011-0901-2>
- Paul J, Lim WM, O’Cass A, Hao AW, Bresciani S (2021) Scientific procedures and rationales for systematic literature reviews (SPAR-4-SLR). *Int J Consum Stud* 45(4):O1–O16
- Pendse MK, Nerlekar VS, Darda P (2023) A comprehensive look at greenwashing from 1996 to 2021: a bibliometric analysis. *J Indian Bus Res* 15(1):157–186. <https://doi.org/10.1108/JIBR-04-2022-0115>
- Peng X, Li J, Tang Q, Lan YC, Cui X (2023) Do environmental scores become multinational corporations’ strategic “greenwashing” tool for window-dressing carbon reduction? A cross-cultural analysis. *Bus Strateg Environ*. <https://doi.org/10.1002/bse.3586>
- *Policarpo MC, Apaolaza V, Hartmann P, Paredes MR, D’Souza C (2023) Social cynicism, greenwashing, and trust in green clothing brands. *Int J Consum Stud* 47(5):1950–1961. <https://doi.org/10.1111/ijc.s.12971>
- *Pope S, Wæraas A (2016) CSR-washing is rare: a conceptual framework, literature review, and critique. *J Bus Ethics* 137(1):173–193. <https://doi.org/10.1007/s10551-015-2546-z>
- *Rahman SU, Nguyen-Viet B (2023) Towards sustainable development: coupling green marketing strategies and consumer perceptions in addressing greenwashing. *Bus Strateg Environ* 32(4):2420–2433. <https://doi.org/10.1002/bse.3256>
- *Rahman SA, Amran A, Ahmad NH, Taghizadeh SK (2015) Supporting entrepreneurial business success at the base of pyramid through entrepreneurial competencies. *Manag Decis* 53(6):1203–1223. <https://doi.org/10.1108/MD-08-2014-0531>
- RepRisk (2023) On the rise: navigating the wave of greenwashing and social washing. <https://www.repris.k.com/research-insights/reports/>. Accessed 28 December 2023
- *Robertson JL, Montgomery AW, Ozbilir T (2023) Employees’ response to corporate greenwashing. *Bus Strateg Environ* 32(7):4015–4027. <https://doi.org/10.1002/bse.3351>
- *Roulet TJ, Touboul S (2015) The intentions with which the road is paved: attitudes to liberalism as determinants of greenwashing. *J Bus Ethics* 128(2):305–320. <https://doi.org/10.1007/s10551-014-2097-8>
- Santos C, Coelho A, Marques A (2023) A systematic literature review on greenwashing and its relationship to stakeholders: state of art and future research agenda. *Manag Rev Q*. <https://doi.org/10.1007/s11301-023-00337-5>
- Sauer PC, Seuring S (2023) How to conduct systematic literature reviews in management research: a guide in 6 steps and 14 decisions. *RMS* 17(5):1899–1933
- *Schmuck D, Matthes J, Naderer B (2018) Misleading consumers with green advertising? An affect–reason–involvement account of greenwashing effects in environmental advertising. *J Advert* 47(2):127–145. <https://doi.org/10.1080/00913367.2018.1452652>
- Schöbel S, Schmitt A, Benner D, Saqr M, Janson A, Leimeister JM (2024) Charting the evolution and future of conversational agents: a research agenda along five waves and new frontiers. *Inf Syst Front* 26(2):729–754
- Schwaeke J, Gerlich C, Nguyen HL, Kanbach DK, Gast J (2025) Artificial intelligence (AI) for good? Enabling organizational change towards sustainability. *Rev Manag Sci*. <https://doi.org/10.1007/s11846-025-00840-x>
- Sciulli N, Adhariani D (2023) The use of integrated reports to enhance stakeholder engagement. *J Account Organ Chang* 19(3):447–473. <https://doi.org/10.1108/JAOC-11-2021-0156>

- *Seele P, Gatti L (2017) Greenwashing revisited: in search of a typology and accusation-based definition incorporating legitimacy strategies. *Bus Strateg Environ* 26(2):239–252. <https://doi.org/10.1002/bsce.1912>
- *Seele P, Schultz MD (2022) From greenwashing to machinewashing: a model and future directions derived from reasoning by analogy. *J Bus Ethics* 178(4):1063–1089. <https://doi.org/10.1007/s10551-022-05054-9>
- *Shi X, Ma J, Jiang A, Wei S, Yue L (2023) Green bonds: Green investments or greenwashing? *Int Rev Financ Anal*. <https://doi.org/10.1016/j.irfa.2023.102850>
- Shuang Q, Lu J, Wang S, Callari D, Cucari N (2024) Exploring the power of informal institutions: How does social trust affect corporate's greenwashing strategy. *Bus Strateg Environ*. <https://doi.org/10.1002/bse.3964>
- *Siano A, Vollero A, Conte F, Amabile S (2017) “More than words”: expanding the taxonomy of greenwashing after the Volkswagen scandal. *J Bus Res* 71:27–37. <https://doi.org/10.1016/j.jbusres.2016.11.002>
- *Sterbenk Y, Champlin S, Windels K, Shelton S (2022) Is femvertising the new greenwashing? Examining corporate commitment to gender equality. *J Bus Ethics* 177(3):491–505. <https://doi.org/10.1007/s10551-021-04755-x>
- *Strauß N, Krakow J, Chesney M (2023) It's the news, stupid! the relationship between news attention, literacy, trust, greenwashing perceptions, and sustainable finance investment in Switzerland. *J Sustain Financ Invest* 13(4):1480–1505. <https://doi.org/10.1080/20430795.2023.2226792>
- *Szabo S, Webster J (2021) Perceived greenwashing: the effects of green marketing on environmental and product perceptions. *J Bus Ethics* 171(4):719–739. <https://doi.org/10.1007/s10551-020-04461-0>
- Talpur S, Nadeem M, Roberts H (2023) Corporate social responsibility decoupling: a systematic literature review and future research agenda. *J Appl Account Res*. <https://doi.org/10.1108/JAAR-08-2022-0223>
- Tashman P, Marano V, Kostova T (2019) Walking the walk or talking the talk? Corporate social responsibility decoupling in emerging market multinationals. *J Int Bus Stud* 50(2):153–171. <https://doi.org/10.1057/s41267-018-0171-7>
- *Testa F, Boiral O, Iraldo F (2018a) Internalization of environmental practices and institutional complexity: Can stakeholders pressures encourage greenwashing? *J Bus Ethics* 147(2):287–307. <https://doi.org/10.1007/s10551-015-2960-2>
- *Testa F, Miroshnichenko I, Barontini R, Frey M (2018b) Does it pay to be a greenwasher or a brown-washer? *Bus Strateg Environ* 27(7):1104–1116. <https://doi.org/10.1002/bse.2058>
- *Testa F, Iovino R, Iraldo F (2020) The circular economy and consumer behaviour: the mediating role of information seeking in buying circular packaging. *Bus Strateg Environ* 29(8):3435–3448. <https://doi.org/10.1002/bse.2587>
- Tilba A (2022) Appearance or substance of Stewardship and ESG reporting? The challenges of translating ‘commitment’ into tangible outcomes. *Sustain Account Manag Policy J* 13(5):1015–1032. <https://doi.org/10.1108/SAMPJ-03-2021-0091>
- *Topal İ, Nart S, Akar C, Erkollar A (2020) The effect of greenwashing on online consumer engagement: a comparative study in France, Germany, Turkey, and the United Kingdom. *Bus Strateg Environ* 29(2):465–480. <https://doi.org/10.1002/bse.2380>
- *Torelli R, Balluchi F, Lazzini A (2020) Greenwashing and environmental communication: effects on stakeholders' perceptions. *Bus Strateg Environ* 29(2):407–421. <https://doi.org/10.1002/bse.2373>
- Tranfield D, Denyer D, Smart P (2003) Towards a methodology for developing evidence-informed management knowledge by means of systematic review. *Br J Manag* 14(3):207–222
- Tuhkanen H, Vulturius G (2022) Are green bonds funding the transition? Investigating the link between companies' climate targets and green debt financing. *J Sustain Financ Invest* 12(4):1194–1216. <https://doi.org/10.1080/20430795.2020.1857634>
- Vangeli A, Małecka A, Mitrega M, Pfajfar G (2023) From greenwashing to green B2B marketing: a systematic literature review. *Ind Mark Manag* 115:281–299. <https://doi.org/10.1016/j.indmarman.2023.10.002>
- Velte P (2023) Determinants and consequences of corporate social responsibility decoupling—status quo and limitations of recent empirical quantitative research. *Corp Soc Responsib Environ Manag* 30(6):2695–2717. <https://doi.org/10.1002/csr.2538>
- Verborgh R, De Wilde M (2013) Using openrefine. Packt Publishing Ltd
- Vrontis D, Christofi M, Pereira V, Tarba S, Makrides A, Trichina E (2022) Artificial intelligence, robotics, advanced technologies and human resource management: a systematic review. *Int J Hum Resour Manag* 33(6):1237–1266

- Wagner T, Korschun D, Troebbs CC (2020) Deconstructing corporate hypocrisy: a delineation of its behavioral, moral, and attributional facets. *J Bus Res* 114:385–394
- Wang W, Ma D, Wu F, Sun M, Xu S, Hua Q, Sun Z (2023) Exploring the knowledge structure and hotspot evolution of greenwashing: a visual analysis based on bibliometrics. *Sustainability* 15(3):2290
- *Wu MW, Shen CH (2013) Corporate social responsibility in the banking industry: motives and financial performance. *J Bank Financ* 37(9):3529–3547
- *Wu Y, Zhang K, Xie J (2020) Bad greenwashing, good greenwashing: corporate social responsibility and information transparency. *Manag Sci* 66(7):3095–3112. <https://doi.org/10.1287/mnsc.2019.3340>
- *Yamoah FA, Sivarajah U, Mahroof K, Peña IG (2022) Demystifying corporate inertia towards transition to circular economy: a management frame of reference. *Int J Prod Econ*. <https://doi.org/10.1016/j.jpe.2021.108388>
- Yang Z, Nguyen TTH, Nguyen HN, Nguyen TTN, Cao TT (2020) Greenwashing behaviours: causes, taxonomy and consequences based on a systematic literature review. *J Bus Econ Manag* 21(5):1486–1507. <https://doi.org/10.3846/jbem.2020.13225>
- Yang H, Shi X, Bhutto MY, Ertz M (2024) Do corporate social responsibility and technological innovation get along? A systematic review and future research agenda. *J Innov Knowl* 9(1):100462
- Ye F, Jiang S, Li Y, Tan KH, Kumar A (2022) The value and pitfall of CM's sustainability signal considering OEM's greenwash. *IEEE Trans Eng Manag*. <https://doi.org/10.1109/TEM.2022.3199788>
- *Zhang W, Qin C, Zhang W (2023) Top management team characteristics, technological innovation and firm's greenwashing: evidence from China's heavy-polluting industries. *Technol Forecast Soc Change*. <https://doi.org/10.1016/j.techfore.2023.122522>
- Zioło M, Bąk I, Spoz A (2024) Literature review of greenwashing research: state of the art. *Corp Soc Responsib Environ Manag*. <https://doi.org/10.1002/csr.2842>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Canio Forliano¹  · Enrico Battisti¹ · Paola de Bernardi¹ · Tomáš Klietík²

✉ Canio Forliano
canio.forliano@unito.it

Enrico Battisti
enrico.battisti@unito.it

Paola de Bernardi
paola.debernardi@unito.it

Tomáš Klietík
tomas.klietik@fpedas.uniza.sk

¹ Department of Management, University of Turin, Turin, Italy

² Department of Finance and Financial Management, University of Zilina, Zilina, Slovakia